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Question 1

Not answered

Marked out of 3.00

Event Rates in Clinical Trials

Lead In: Which of the following terms best describe the concepts below?

1. Experimental Event Rate - Control Event Rate 
2. $1 / (\text{Experimental Event Rate} - \text{Control Event Rate})$ 
3. Proportion of experimental event/proportion of control event 

Explanation: 1. The absolute arithmetic difference in rates of observed outcomes of interest between experimental and control participants in a trial, calculated as $|\text{EER} - \text{CER}|$ is the risk difference. It is also called the ARR or ABI (Absolute Benefit Increase) = $\text{CER} (\text{Control Event Rate}) - \text{EER} (\text{Experimental Event Rate})$.

2. The Number Needed to Treat (NNT) is the number of patients you need to treat to prevent one additional bad outcome (death, stroke, etc.). To calculate the NNT, you need to know the Absolute Risk Reduction (ARR); the NNT is the inverse of the ARR: $\text{NNT} = 1/\text{ARR}$, NNTs are always rounded up to the nearest whole number.

3. The ratio of risk in the treated group (EER) to the risk in the control group (CER) is called Relative Risk or Risk Ratio (RR). This is often used in RCTs and cohort studies and is calculated as EER/CER .

Note: An odds ratio (OR) is a measure of association between an exposure and an outcome. The OR represents the odds (not proportions) that an outcome will occur given a particular exposure, compared to the odds of the outcome occurring in the absence of that exposure.

Ref: <https://ebm-tools.knowledgetranslation.net/resource/glossary>

Question 2

Not answered

Marked out of 3.00

EMI- Variations in cognitive impairment

Lead In: Which of the following is the most likely diagnosis in each of the below scenarios?

1. Mr. Frank Lewis is a 41-year-old man who has presented to his GP with a history of memory decline for several months, unsteady gait and recurrent falls. On examination he had nystagmus, dysarthria and lower limb weakness; his reflexes were absent. His father died in his 30's from a myocardial infarction ❌

2. Mrs. Deborah Banner is a 60-year-old woman with a history of Type 2 diabetes mellitus and hyperlipidemia. Her husband is concerned, as her memory appears to have declined gradually over a year. He persuades her to go to the GP but she appears unconcerned about his worries. ❌

3. Mr. Charles Rawlins is a 71-year-old man who has been under follow-up with neurology for the last few years for persistent rigidity and tremors. He has now presented with visual hallucinations, that he finds distressing, and intermittent confusion.

 ❌

Explanation: 1) Idiopathic normal pressure hydrocephalus is characterised by dementia, gait ataxia, urinary incontinence, recurrent falls and enlarged cerebral ventricles. It is associated with an increased prevalence of cardiovascular diseases. 2) Binswanger's disease is a form of small vessel vascular dementia caused by damage to the white brain matter. It is characterized by loss of memory and intellectual function and by changes in mood and affect. 3) Lewy body dementia is an umbrella term that includes Parkinson's disease dementia (PDD) and dementia with Lewy bodies (DLB), two dementias characterized by abnormal deposits of the protein alpha-synuclein in the brain. Dementia with Lewy bodies is distinguished from Parkinson's disease dementia by the time frame in which dementia symptoms appear relative to parkinsonian symptoms. DLB is diagnosed when cognitive symptoms begin before or at the same time as Parkinsonism, while PDD is the diagnosis when Parkinson's disease is well established before dementia occurs.

Question 3

Not answered

Marked out of 3.00

Theme-Neurotic and Somatic Symptom Disorders

Lead In: Which of the following is the most likely diagnosis in each of the scenarios given below?

1. Miss Sylvia Young is a 32-year-old woman who is convinced that she is suffering from a severe autoimmune disease. She has attended numerous doctors, both within the NHS and in private sector, seeking a diagnosis but has been advised that no sign of an autoimmune disorder can be found. She is very upset with the health system now and continues to be convinced that the doctors are missing a serious health concern. ❌

2. Mr Giles Hardy is a 52-year-old man who is the primary caregiver for his mother with Alzheimer's disease. He has contacted the ambulance service to report developing sudden onset weakness in his right leg. He is admitted urgently to the hospital. A thorough physical examination and investigations done while he was clerked in indicates no evidence of any neurological abnormality. ❌

3. Mr. Philip Crawford is a 27 year old man who believes that his nose is abnormally large. He has had an expensive rhinoplasty arranged privately but remains convinced that his nose is unusually shaped even after the surgical correction. The operating cosmetic surgeon showed him multiple pictures explaining the anatomy of nose in comparison to the pictures of his own nose. Unconvinced, he has now requested another plastic surgery and the surgeon declined this request and referred him to the local mental health team. ❌

Explanation: 1) Hypochondriasis is characterised by the individual being convinced that they are suffering from a significant illness and then seeking a specific diagnosis. They may experience some symptoms, but unlike in somatisation disorder, the focus is more on the diagnosis. 2) Conversion disorder presents with neurological symptoms, such as numbness, blindness, paralysis or fits, which are not consistent with a well-established organic cause, which cause significant distress, and can be traced back to a psychological trigger. 3) Body dysmorphic disorder, or dysmorphophobia, is a mental disorder characterised by the obsessive idea that some aspect of one's own body or appearance is severely flawed. Affected individuals will attempt to conceal or rectify this perceived abnormality, often via plastic surgery or similar procedures. This does not usually address the problem and they often present again with the same complaint.

Question 4

Not answered

Marked out of 1.00

Mr Stephen Williams is a 42-year-old man with a history of alcohol dependence who has presented to an emergency dentist with a dental abscess. The tooth is removed but he requires an oral antibiotic. Which of the following is contraindicated in patients with alcohol dependence?

Select one:

- ☐ Doxycycline
- ☐ Metronidazole
- ☐ Amoxicillin
- ☐ Trimethoprim
- ☐ Ciprofloxacin

Your answer is incorrect.

Metronidazole, an effective antianaerobic agent, has been associated with a disulfiram-like adverse reaction when it is taken around the same time as alcohol. This is due to the blocking of hepatic aldehyde dehydrogenase (ALDH) enzyme followed by the accumulation of acetaldehyde in the blood. This has been questioned in recent years, in terms of whether it is a true reaction or if only a proportion of the population is vulnerable to it, but it remains to be considered a contraindication. Ref: Visapää JP et al. Lack of disulfiram-like reaction with metronidazole and ethanol. *Annals of Pharmacotherapy*. 2002 Jun;36(6):971-4.

The correct answer is: Metronidazole

Question 5

Not answered

Marked out of 1.00

Mr Adams is a 34-year-old man who has recently been diagnosed with pathological gambling disorder, with him having marked difficulties with the use of online gambling. Which of the following statement is true for pathological gambling disorder?

Select one:

- ☐ There is no evidence for mood-stabilising medication
- ☐ Motivational interviewing showed no difference compared to placebo
- ☐ SSRIs are associated with favourable outcomes in patients with no comorbid psychiatric disorder
- ☐ Naltrexone can be used as last resort in patient who has failed other treatments
- ☐ There is little evidence for the use of CBT

Your answer is incorrect.

There is evidence for the use of naltrexone, although it seems to be more effective for more severe gambling disorders and can be associated with increases in liver enzymes and marked nausea. SSRIs have some evidence for effectiveness, both in individuals with comorbid disorders and those without. Motivational interviewing and CBT have some evidence for effectiveness. Lithium has some preliminary evidence for efficacy. Ref: Goslar M, Leibetseder M, Muench HM, Hofmann SG, Laireiter AR. Pharmacological treatments for disordered gambling: a meta-analysis. Journal of gambling studies. 2019 Jun 15;35(2):415-45.

The correct answer is: Naltrexone can be used as last resort in patient who has failed other treatments

Question 6

Not answered

Marked out of 1.00

Mr David Young is a 38-year-old man with a long-standing opiate dependence. He has recently suffered a relapse and has been using heroin again and has rapidly increased the amount he is using. He reports that he still feels the need to use more. Which of the following opioid effects do not usually show any tolerance effects?

Select one:

- ☐ Euphoria
- ☐ Nausea
- ☐ Respiratory depression
- ☐ Tachycardia
- ☐ Miosis

Your answer is incorrect.

Despite tolerance to the other effects of opioids, such as elation, nausea and respiratory depression occurring relatively rapidly, there is little to no tolerance to the effects of constipation and miosis.

The correct answer is: Miosis

Question 7

Not answered

Marked out of 1.00

Mr Gary Cleaver is a 42-year-old man who has just completed a opioid detoxification and has remained abstinent from all opiates for the last 2 weeks. His wife is very supportive and his social situation is relatively stable. Which of the following would be most appropriate for him now?

Select one:

- ☐ Acamprosate
- ☐ Naltrexone
- ☐ Buprenorphine
- ☐ Inpatient rehabilitation
- ☐ Community rehabilitation

Your answer is incorrect.

Naltrexone is an opioid receptor antagonist that is indicated for use as an adjunct to aid formerly opiate dependent patients who remain abstinent now. It blocks the euphoric effects of opiates but patients must also have been abstinent for 7-10 days prior to prevent it precipitating severe withdrawals.

The correct answer is: Naltrexone

Question 8

Not answered

Marked out of 1.00

Max is a 44-year-old male. He has mild alcohol dependence without significant physical sequelae. He attends his first appointment at the local drug and alcohol service with his partner. He asks for pharmacological intervention to reduce his cravings for alcohol. Which of the following treatment combinations should you offer him?

Select one:

- ☐ Methadone and behavioural couple therapy
- ☐ Acamprosate and behavioural couple therapy
- ☐ Buprenorphine and psychodynamic therapy
- ☐ Acamprosate and psychodynamic therapy
- ☐ Naltrexone and, acceptance and commitment therapy

Your answer is incorrect.

Acamprosate works by enhancing GABA transmission in the brain thus restoring GABA imbalance seen in alcohol dependence. Patients who have taken it report a suppressed 'urge to drink.' Acamprosate and CBT is more effective than acamprosate or CBT alone. Alcohol-focused Behavioral Couple Therapy (ABCT) is a CBT model that works to identify and manage the multiple factors that can influence a person's drinking, including individual, familial and other environmental variables Ref: Shen WW. Anticraving therapy for alcohol use disorder: A clinical review. Neuropsychopharmacology reports. 2018 Sep;38(3):105-16.

The correct answer is: Acamprosate and behavioural couple therapy

Question 9

Not answered

Marked out of 1.00

In a general hospital ward you review Richard, a 45-year-old gentleman with a harmful alcohol habit. When you ask Richard about his views on his alcohol consumption, he tells you that he understands the harm caused by alcohol and agrees upon making a change. Which stage of alcohol recovery is Richard in?

Select one:

- ☐ Maintenance stage
- ☐ Preparation stage
- ☐ Precontemplation stage
- ☐ Contemplation stage
- ☐ Action stage

Your answer is incorrect.

There are five stages of alcohol recovery: precontemplation, contemplation, preparation, action and maintenance / recovery. An individual in the preparation stage is aware of the consequence(s) of their harmful behaviour and makes a commitment to change. Ref: <https://www.ncbi.nlm.nih.gov/books/NBK64968/>

The correct answer is: Preparation stage

Question 10

Not answered

Marked out of 1.00

In your addictions clinic, you see Clive, a 56-year-old former professional footballer, with a history of alcohol dependence. He has been sober for five years but is worried about becoming dependent on alcohol once again. Since his divorce, he has found it difficult to cope with stress. He is concerned about the onset of craving and asks you whether there is any medication he can take to suppress these urges. What do you suggest?

Select one:

- ☐ Mirtazapine
- ☐ Acamprosate
- ☐ Disulfiram
- ☐ Venlafaxine
- ☐ Alprazolam

Your answer is incorrect.

Acamprosate works by enhancing GABA transmission in the brain thus restoring GABA imbalance seen in alcohol dependence. Patients who have taken it report a suppressed 'urge to drink.' Acamprosate and CBT is more effective than acamprosate or CBT alone.

The correct answer is: Acamprosate

Question **11**

Not answered

Marked out of 1.00

Gary is a 51-year-old male with a history of alcohol dependence. You are asked by the gastroenterologist to assess him on the medical ward. On assessment, you learn that Gary has experienced a number of falls due to losing his balance. He complains of blurred vision. You also notice that he appears confused. Based on his symptoms, which medication would you treat Gary with?

Select one:

- ☐ Thiamine
- ☐ Alprazolam
- ☐ Disulfiram
- ☐ Carbamazepine
- ☐ Acamprosate

Your answer is incorrect.

Wernicke's encephalopathy (WE) is an acute condition characterised by a well-known triad of ataxia, ophthalmoplegia and confusion. It is caused by thiamine deficiency usually related to alcohol abuse. WE is treated with thiamine. Thiamine should be given before any IV glucose as glucose infusion exacerbates thiamine deficiency biochemically. WE is initially reversible but, if untreated or treated with inadequate thiamine replenishment, it can result in irreversible Korsakoff's syndrome in 84% of survivors, and 5% death rate. Ref: Day E, et al. Thiamine for prevention and treatment of Wernicke-Korsakoff Syndrome in people who abuse alcohol. Cochrane Database of Systematic Reviews. 2013(7).

The correct answer is: Thiamine

Question 12

Not answered

Marked out of 1.00

Kate Forlock is a 2-year-old girl who was adopted into her current family around 9 months ago. Her adoptive mother has raised concerns that Kate does not seem to respond to affection, does not seek out cuddles and seems to avoid eye contact at times. She plays imaginatively and her language skills are developing relatively normally. Which of the following is the most likely diagnosis?

Select one:

- ☐ ADHD
- ☐ ASD
- ☐ Reactive attachment disorder
- ☐ Social communication disorder
- ☐ Auditory processing disorder

Your answer is incorrect.

Reactive attachment disorder is characterized by markedly disturbed and developmentally inappropriate ways of relating socially in most contexts. It can take the form of a persistent failure to initiate or respond to most social interactions in a developmentally appropriate way. It arises from a failure to form normal attachments to primary caregivers in early childhood. Such a failure could result from severe early experiences of neglect, abuse, abrupt separation from caregivers between the ages of six months and three years, frequent change of caregivers, or a lack of caregiver responsiveness to a child's communicative efforts.

The correct answer is: Reactive attachment disorder

Question **13**

Not answered

Marked out of 1.00

Holly Ward is a 9-year-old girl whose teachers have raised concerns about her. She has a good vocabulary but cannot write sentences of longer than 6 words or read books, which would normally be suitable for children 3-4 years, her junior. She has started to have angry outbursts in class. On interview, she presents as bright and is able to answer your questions clearly. Which of the following is the most likely diagnosis?

Select one:

- ☐ Social communication disorder
- ☐ ADHD
- ☐ Dyslexia
- ☐ Oppositional defiant disorder
- ☐ ASD

Your answer is incorrect.

Dyslexia, also known as reading disorder, is characterized by trouble with reading despite normal intelligence. Problems may include difficulties in spelling words, reading quickly, writing words, "sounding out" words in the head, pronouncing words when reading aloud and understanding what one reads.

The correct answer is: Dyslexia

Question **14**

Not answered

Marked out of 1.00

Simon Lovell is a 10-year-old boy who struggles to concentrate at school and becomes upset in loud environments. However, he is happy at home and can produce excellent work at home without difficulty. Which of the following is the most likely diagnosis?

Select one:

- ☐ Auditory processing disorder
- ☐ ASD
- ☐ Dyslexia
- ☐ ADHD
- ☐ Social communication disorder

Your answer is incorrect.

Auditory processing disorder is a disorder that affects the way the brain processes auditory information. Individuals usually have normal structure and function of the outer, middle, and inner ear. However, they cannot process the information they hear in the same way as others do, which leads to difficulties in recognizing and interpreting sounds, especially the sounds composing speech.

The correct answer is: Auditory processing disorder

Question 15

Not answered

Marked out of 1.00

Miss Elisa Carron is a 24-year-old woman who has recently been diagnosed with ADHD of moderate severity. Which of the following would be the most appropriate initial treatment?

Select one:

- ☐ Social skills training
- ☐ Atomoxetine
- ☐ Methylphenidate
- ☐ CBT
- ☐ Clonidine

Your answer is incorrect.

NICE recommends that adults with ADHD should be offered methylphenidate or dexamfetamine as a first line choice, trialing the other if the first is not effective. Atomoxetine can then be considered. Non-pharmacological options are only appropriate if the patient cannot tolerate medication, does not take it regularly, or if they have chosen to go without.

The correct answer is: Methylphenidate

Question 16

Not answered

Marked out of 1.00

Keiran Fowles is a 12-year-old boy with diagnoses of ASD and generalised anxiety. He and his family are referred to the team's family therapist. They inform the family that they may use paradoxical injunction during the course of the therapy. In which of the following forms of family therapy would this be most likely to be used?

Select one:

- ☐ Systemic family therapy
- ☐ Strategic family therapy
- ☐ Dynamic family therapy
- ☐ Structural family therapy
- ☐ Narrative family therapy

Your answer is incorrect.

Paradoxical techniques such as paradoxical injunction were originally introduced as, and mostly used in, systemic therapies. But these interventions cut across theoretical boundaries and are "used by virtually all schools of psychotherapy", including strategic family therapy. These techniques include counterintuitive suggestions, for example, suggesting the family to take action seemingly in opposition to their therapeutic goals.

Ref: Shoham, V. and Rohrbaugh, M.J. (2010). Paradoxical Intervention. In The Corsini Encyclopedia of Psychology (eds I.B. Weiner and W.E. Craighead).

The correct answer is: Systemic family therapy

Question 17

Not answered

Marked out of 1.00

Eve Mitchell is a 3-year girl who is learning to brush her teeth. Her parents and older siblings sit and brush their teeth with her in the morning and she is copying them and gradually improving. Which of the following describes this form of learning?

Select one:

- ☐ Social
- ☐ Habituation
- ☐ Operant
- ☐ Imprinting
- ☐ Classical

Your answer is incorrect.

Social Learning Theory is a theory of learning process and social behavior, which proposes that new behaviours can be acquired by observing and imitating others.

The correct answer is: Social

Question 18

Not answered

Marked out of 1.00

Yara Swanson is a 9-year-old girl with ADHD. She has difficulty focusing on homework at school and the psychologist in the CAMHS team has suggested that they use a reinforcement method to encourage her, in addition to medication therapy. They suggest that they do not reinforce each and every positive behaviour but only some. Which of the following would best describe this approach?

Select one:

- ☐ Continuous reinforcement
- ☐ Contingent reinforcement
- ☐ Variable ratio reinforcement
- ☐ Non-contingent reinforcement
- ☐ Differential reinforcement

Your answer is incorrect.

Variable ratio reinforcement involves a positive reinforcer being provided after an average number of positive behaviours, with the exact number between reinforcement varying each time. In fixed ratio reinforcement, positive behaviours are rewarded every 3rd or 4th time, or whatever interval is being used. Contingent reinforcement is that which is dependent upon a positive behaviour being performed. In non-contingent reinforcement, a reward is provided regardless of whether there has been positive behaviour. Differential reinforcement is the process of reinforcing only the appropriate response (or behaviour you wish to increase) and applying extinction to all other responses.

The correct answer is: Variable ratio reinforcement

Question 19

Not answered

Marked out of 1.00

A four-year-old boy is granted 30 minutes of screen time to watch his favourite cartoon on the provision that he finishes eating his dinner. Which category of operant conditioning does this fall into?

Select one:

- ☐ Extinction
- ☐ Negative reinforcement
- ☐ Negative punishment
- ☐ Positive reinforcement
- ☐ Positive punishment

Your answer is incorrect.

Operant conditioning is a method of learning that occurs through rewards and punishments for behaviour. An individual will make an association between a particular behaviour and a consequence (Skinner, 1938). The above scenario describes positive reinforcement, whereby a behaviour (response) is followed by another stimulus that is rewarding, which serves to increase the frequency of the behaviour.

The correct answer is: Positive reinforcement

Question **20**

Not answered

Marked out of 1.00

This term refers to the range of tasks that children cannot yet accomplish on their own but can do with the help of adults or other children.

Select one:

- ☐ Target problem procedure
- ☐ Negative reinforcement
- ☐ Target setting
- ☐ Positive reinforcement
- ☐ Zone of proximal development

Your answer is incorrect.

Vygotsky's zone of proximal development (ZPD) refers to the range of tasks that children cannot yet accomplish on their own but can do with the help of adults or other children. ZPD functions are not fully achieved yet but are in the process / 'pipeline.' Whilst actual achievement measures development after completion retrospectively, ZPD has a prospective sense as to what the child will be achieving in the future.

The correct answer is: Zone of proximal development

Question **21**

Not answered

Marked out of 1.00

Mark Jacobs is a 10-year-old boy who has difficulties with recurrent self-injurious behaviour. He has features of abnormally increased distance between his eyes and a severe intellectual impairment. His mother recalls that he was 'supple' shortly after birth. Which of the following is the most likely diagnosis?

Select one:

- ☐ Prader Willi syndrome
- ☐ Fragile X syndrome
- ☐ Cri du chat syndrome
- ☐ Angelman syndrome
- ☐ Di George syndrome

Your answer is incorrect.

Cri du chat syndrome, is a rare genetic disorder due to chromosome deletion on chromosome 5. Apart from the characteristic hypertelorism and abnormal cry after birth, it can be associated with self-injurious behaviour and challenging behaviour later in life. Ref: <https://rarediseases.info.nih.gov/diseases/6213/cri-du-chat-syndrome>

The correct answer is: Cri du chat syndrome

Question **22**

Not answered

Marked out of 1.00

Iain Graves is an 8-year-old boy with a diagnosis of ADHD. He has recently started methylphenidate and it appears to be helping him at school. His parents want to know how it works and ask about the neural circuits involved in the symptoms. Which of the following neural pathways are implicated in ADHD?

Select one:

- ☐ Corticobulbar
- ☐ Corticostriatal
- ☐ Arcuate fasciculus
- ☐ Frontostriatal
- ☐ Nigrostriatal

Your answer is incorrect.

Recent fMRI studies using inhibitory control, working memory, and attentional tasks in individuals with ADHD and healthy comparison subjects have shown underactivation of fronto-striatal, fronto-parietal, and ventral attention networks in those with ADHD. Structural imaging has revealed significantly smaller brains in children compared to unaffected controls. The prefrontal cortex, basal ganglia and cerebellum are differentially affected and evidence indicating reduced connectivity in white matter tracts in key brain areas is emerging. Growing evidence points to the involvement of the frontostriatal network as a likely contributor to the pathophysiology of ADHD (Ref: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4437560/>).

The correct answer is: Frontostriatal

Question **23**

Not answered

Marked out of 1.00

Mr Graham Harris is a 26-year-old man with a diagnosis of emotionally unstable and paranoid personality disorder who has recently been imprisoned for ABH. Which of the following is the closest prevalence of personality disorder among convicted male prisoner?

Select one:

- ☐ 40%
- ☐ 30%
- ☐ 50%
- ☐ 65%
- ☐ 80%

Your answer is incorrect.

Around 65% of male prisoners have a personality disorder, with 47% meeting criteria for dissocial personality disorder. (Ref: Fazel S, and Danesh J. Serious mental disorder in 23?000 prisoners: a systematic review of 62 surveys. The Lancet, 359 (9306), 2002Feb16, pp545-550).

The correct answer is: 65%

Question **24**

Not answered

Marked out of 1.00

Teagan is a 24-year-old medical student with a history of depression. She has been taking paroxetine for several months and feels this has helped with the remission of her illness. Teagan often forgets to take her medication with her when going on weekend trips. She presents to A&E after one such occasion of missing her medication, and you are asked to see her. She presents as irritable and complains of nausea, dizziness, vomiting and nightmares. What treatment do you recommend for her?

Select one:

- ☐ Commence another SSRI antidepressant
- ☐ Commence diazepam
- ☐ Recommence her prescribed dose of paroxetine and gradually reduce
- ☐ Do nothing
- ☐ Double her prescribed dose of paroxetine

Your answer is incorrect.

Paroxetine is an SSRI antidepressant. Being one of the SSRIs with the shortest half-life, stopping paroxetine abruptly may result in one or more of the following withdrawal symptoms: irritability, nausea, feeling dizzy, vomiting, nightmares, headache, and/or paresthesia. Recommencing the prescribed dose of paroxetine and gradually decreasing the dose is one of the best ways to manage and reduce symptoms of discontinuation.

Ref: Wilson E, Lader M. A review of the management of antidepressant discontinuation symptoms. Therapeutic advances in psychopharmacology. 2015 Dec;5(6):357-68.

The correct answer is: Recommence her prescribed dose of paroxetine and gradually reduce

Question 25

Not answered

Marked out of 1.00

Mr Travers is a 57-year-old man who has presented to his local A&E with sudden onset paralysis affecting his right leg. During admission it is noted that he appears to shift his right leg in bed and it is suspected that he may be experiencing conversion symptoms. Which of the following tests could be best to clarify his diagnosis?

Select one:

- ☐ EEG
- ☐ Hoover test
- ☐ MRI head
- ☐ Mental State Examination
- ☐ Reflex testing

Your answer is incorrect.

The Hoover test is designed to elicit Hoover's sign. It is based on the principle of crossed extensor reflex. The Hoover test aims to distinguish organic from non-organic leg paresis. In patients with organic paresis the assessor should feel the 'normal' limb's heel push down against his or her hand as the patient attempts to raise the 'weak' leg's hip. If the normal leg's heel pushing down isn't felt, this suggests a functional weakness, which is also known as conversion disorder.

Ref: Stone J, et al. Functional symptoms and signs in neurology: assessment and diagnosis. Journal of Neurology, Neurosurgery & Psychiatry. 2005 Mar 1;76(suppl 1):i2-12.

The correct answer is: Hoover test

Question **26**

Not answered

Marked out of 1.00

Miss Eve Campbell is a 19-year-old woman who has been brought to A&E by her family after she has self-harmed after a relationship has come to an end. Her parents are very worried about whether she could harm herself more seriously in the future. Which of the following best describes the probability of a person completing suicide following an act of self-harm in the preceding year?

Select one:

- ☐ 0.1
- ☐ 1/100000
- ☐ 1/10000
- ☐ 1/1000
- ☐ 1/100

Your answer is incorrect.

The risk of suicide within a year of self-harm is around 1%, with this rising to 5% over 10 years. (Ref: Chang B, Gitlin D, Patel R. The depressed patient and suicidal patient in the emergency department: evidence-based management and treatment strategies. *Emergency Medicine Practice*, 2011;Sept, 13(9): 1–23, quiz 23–24. Tintinalli, Judith E. (2010). *Emergency Medicine: A Comprehensive Study Guide* (Emergency Medicine (Tintinalli)). New York: McGraw-Hill Companies. pp. 1940–46).

The correct answer is: 1/100

Question **27**

Not answered

Marked out of 1.00

Mrs Sophie Longworth is a 40-year-old woman who is currently attending a psychologist to address long-term anxiety and relationship difficulties. The psychologist suggests trying 'the empty chair' technique. Which of the following therapies uses this approach?

Select one:

- ☐ Rational emotive therapy
- ☐ Existential therapy
- ☐ DBT
- ☐ Gestalt therapy
- ☐ Interpersonal therapy

Your answer is incorrect.

The 'empty chair' technique is a gestalt therapy technique in which the client engages in a role-played conversation with an imagined person. The patient sits facing an empty chair, and imagines that a family member or some other specific person is sitting there. The patient will talk to the person, either to verbalize things they for some reason would rather not say in real life, or to rehearse future encounters with that person. Ref: Wagner-Moore LE. Gestalt Therapy: Past, Present, Theory, and Research. Psychotherapy: Theory, Research, Practice, Training. 2004;41(2):180.

The correct answer is: Gestalt therapy

Question **28**

Not answered

Marked out of 1.00

Your next work placement will be with the Recovery Team. Which of the following is a guiding principle of the recovery model?

Select one:

- ☐ Ascertaining problems the patient is experiencing
- ☐ Assessing the patient's strengths
- ☐ Undertaking regular psychiatric assessments
- ☐ Ensuring the patient attends their appointments
- ☐ Checking adherence to treatment

Your answer is incorrect.

Recovery teams are community mental health teams for people aged 18-65 years with severe and enduring mental illness, who have complex needs. Aims of the team are to help patients function independently and as well as possible in everyday life within the community. The team uses the recovery model, which emphasizes and supports a person's potential for recovery. The model is strengths-based, which means the focus is on the strengths and resources of the patient rather than their impairment. Ref: Davidson L. Recovery, self management and the expert patient: Changing the culture of mental health from a UK Perspective. J Ment Health. 2005;14:25-35.

The correct answer is: Assessing the patient's strengths

Question 29

Not answered

Marked out of 1.00

Mr Dylan Cook is a 35-year-old man who has a long history of alcohol dependence and has struggled to control his use in the community, despite a period of inpatient rehabilitation. He has attended clinic today and asks whether there is any medication that may help him. Which of the following can be used to manage patient with alcohol dependence?

Select one:

- ☐ Venlafaxine
- ☐ Baclofen
- ☐ Mirtazapine
- ☐ Buspirone
- ☐ Gabapentin

Your answer is incorrect.

There have been some recent studies, which suggest that gabapentin may be effective in alcohol dependence. Baclofen was previously thought to be effective but a recent meta-analysis has cast doubts on this. (Ref: Mason BJ, Quello S, Goodell V, Shadan F, Kyle M, Begovic A. Gabapentin Treatment for Alcohol Dependence: A Randomized Controlled Trial. JAMA Intern Med. 2014 Jan 1; 174(1): 70–77. & Bschor T, Henssler J, Müller M, Baethge C. Baclofen for alcohol use disorder-a systematic meta-analysis. Acta Psychiatr Scand. 2018 Sep;138(3):232–242).

The correct answer is: Gabapentin

Question **30**

Not answered

Marked out of 1.00

Mr Evans is a 26-year-old man who has been brought to his local A&E by his family after becoming increasingly elated and energetic over the last week. He stopped sleeping and started to build a "perpetual motion machine" in his garage. Which of the following would be recommended to treat him?

Select one:

- ☐ Carbamazepine
- ☐ Sodium Valproate
- ☐ Lithium
- ☐ Aripiprazole
- ☐ Olanzapine

Your answer is incorrect.

NICE Bipolar guideline recommends that individuals with a manic episode should be offered haloperidol, olanzapine, quetiapine or risperidone in the first instance, depending upon patient preference, advance directives and clinical context.

The correct answer is: Olanzapine

Question **31**

Not answered

Marked out of 1.00

Mental disorders often are more prevalent in males or females. Amongst children, which of the following has equal prevalence in boys and girls?

Select one:

- ☐ Conduct disorder
- ☐ Tourette's syndrome
- ☐ ASD
- ☐ Trichotillomania
- ☐ OCD

Your answer is incorrect.

OCD, unlike many other anxiety disorders, has equal prevalence in boys and girls. ASD, conduct disorder and Tourette's are more common in boys and trichotillomania is more common in girls. Ref: Boileau B. A review of obsessive-compulsive disorder in children and adolescents. Dialogues in clinical neuroscience. 2011 Dec;13(4):401.

The correct answer is: OCD

Question **32**

Not answered

Marked out of 1.00

Mrs Claire Matthews is a 34-year-old woman who is currently pregnant. She has a son, Arthur, who has a diagnosis of ASD. She wants to know what the likelihood is that her unborn child will also have a diagnosis of ASD. Which of the following is the likelihood that this will occur?

Select one:

- ☐ 15%
- ☐ 15%
- ☐ 50%
- ☐ 25%
- ☐ 1%

Your answer is incorrect.

The baseline likelihood of having a child with ASD is around 1.5%. If a family has one child with ASD, this increases to around 15%. If they have two children with ASD, it increases to around 30%. If the next child is male, they are 2-3 times more likely to have ASD than if they were female.

The correct answer is: 15%

Question **33**

Not answered

Marked out of 1.00

Mr Douglas Andrews is a 32-year-old man with a diagnosis of bipolar affective disorder. He has had a recent episode of mania but has recently recovered and is taking his medication reliably. His driving licence was suspended during his acute illness. He is keen to get back to driving his own car again. How long will his licence remain suspended, now that he has recovered?

Select one:

- ☐ 6 months
- ☐ 3 months
- ☐ 1 month
- ☐ 1 year
- ☐ 2 years

Your answer is incorrect.

The DVLA requires that an individual's license be suspended for 3 months after an episode of mania has resolved, or 6 months if they have had 4 or more episodes of mood disturbance within one year. Patients who have experienced an episode of psychosis and have recovered usually also have their license suspended for 3 months.

The correct answer is: 3 months

Question **34**

Not answered

Marked out of 1.00

26. Miss Thomson is a 24-year-old woman who has been admitted to an inpatient psychiatry unit after beginning to hear voices coming from her light bulbs and having persecutory beliefs about her housemates. Her parents have attended the ward round and are asking what her likely prognosis will be. Which of the following would be the worse prognostic factor in her case?

Select one:

- ☐ Female gender
- ☐ Immigrant
- ☐ Poor compliance with medication
- ☐ Insidious onset
- ☐ Long Duration of Untreated Psychosis

Your answer is incorrect.

Features that predict worse outcome for FEP patients include the longer duration of untreated psychosis (DUP), male gender, non-adherence, poor premorbid adjustment, the insidious mode of onset, the greater severity of negative symptoms, comorbid substance use disorders and having non-affective psychosis. (Ref: Suvisaari J, et al. Is it possible to predict the future in first-episode psychosis?. Frontiers in psychiatry. 2018;9:580.).

The correct answer is: Long Duration of Untreated Psychosis

Question **35**

Not answered

Marked out of 1.00

Which of the following mental illnesses has almost equal prevalence in males and females?

Select one:

- ☐ Hypochondriasis
- ☐ Generalised Anxiety Disorder
- ☐ Bipolar Disorder I
- ☐ PTSD
- ☐ Social Phobia

Your answer is incorrect.

Hypochondriasis is the preoccupation with the fear of having a serious disease, usually one that leads to death or serious disability. This constant preoccupation persists despite negative medical investigations. It is seen equally in men and women, between the ages of 20 and 30 years of age.

The correct answer is: Hypochondriasis

Question **36**

Not answered

Marked out of 1.00

Jane is a 36-year-old female who is an inpatient on a mental health unit. She presents with depressive disorder, current episode: severe. In the past, she has experienced one episode of mania. Which of the following would be the best medication to treat her current episode?

Select one:

- ☐ Quetiapine
- ☐ Sodium valproate
- ☐ Risperidone
- ☐ Lithium
- ☐ Lamotrigine

Your answer is incorrect.

According to diagnostic classification, Jane is suffering from Bipolar Disorder I (occurrence of one or more manic episodes or mixed episodes, with or without a history of one or more depressive episodes). If a person develops moderate or severe bipolar depression and is not taking a drug to treat their bipolar disorder, offer fluoxetine combined with olanzapine, or quetiapine on its own, depending on the person's preference and previous response to treatment (NICE).

The correct answer is: Quetiapine

Question **37**

Not answered

Marked out of 1.00

Which of the following is NOT a symptom of narcolepsy?

Select one:

- ☐ Excessive sleepiness
- ☐ Sleep paralysis
- ☐ Excessive sleepiness
- ☐ Sleep walking
- ☐ Cataplexy

Your answer is incorrect.

Sleepwalking is not part of the classical 'tetrad' of symptoms of narcolepsy that includes excessive sleepiness, hypnagogic hallucinations, sleep paralysis and cataplexy. Excessive sleepiness and associated cataplexy (sudden loss of muscle tone leading to collapse, which is usually preceded by emotional stress) are the most common complaints.

The correct answer is: Sleep walking

Question **38**

Not answered

Marked out of 1.00

Which of the following conditions are most associated with hypochondriasis?

Select one:

- ☐ Depression
- ☐ Panic disorder
- ☐ Obsessive Compulsive Disorder
- ☐ Schizophrenia
- ☐ Generalised Anxiety Disorder

Your answer is incorrect.

In patients with hypochondriasis, anxiety disorders as a group were remarkably common (generalized anxiety disorder 28.5%, panic disorder 14.4%, agoraphobia 16.1%, PTSD and OCD each 12.9%), and despite major depression being the most common single comorbidity, anxiety disorders as a group were more common than depressive disorders as a group. Ref:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4792743/>

The correct answer is: Generalised Anxiety Disorder

Question **39**

Not answered

Marked out of 1.00

Which of the following medical conditions is NOT associated with panic disorder?

Select one:

- ☐ IBD
- ☐ Hypertension
- ☐ COPD
- ☐ Migraine
- ☐ Arthritis

Your answer is incorrect.

Hypertension, migraine, COPD and IBD are common medical comorbidities of panic disorder. Ref: Zaubler TS, Katon W. Panic disorder and medical comorbidity: a review of the medical and psychiatric literature. Bulletin of the Menninger Clinic. 1996 Apr 1;60(2).

The correct answer is: Arthritis

Question 40

Not answered

Marked out of 1.00

In your liaison outpatient clinic, you review Marie who complains of extreme tiredness, aches and pains 'all over' and headaches. She has had these symptoms for years. Her symptoms became worse a few months ago when she started a new job, which she admits has been stressful. Investigations have ruled out a physical illness. Which of the following conditions is the most likely cause of her symptoms?

Select one:

- ☐ Generalised anxiety disorder
- ☐ Bipolar Disorder
- ☐ Depression
- ☐ Munchausen by Proxy
- ☐ Fibromyalgia

Your answer is incorrect.

Fibromyalgia is a long-term condition that causes a number of symptoms including pain all over the body, increased sensitivity to pain, fatigue, muscle stiffness, difficulty sleeping and headaches. The exact cause of the condition is unknown, however it can be triggered by physically or emotionally stressful events. Ref: Häuser W, Fitzcharles MA: Facts and myths pertaining to fibromyalgia. Dialogues Clin Neurosci 2018; 20:53–62

The correct answer is: Fibromyalgia

Question **41**

Not answered

Marked out of 1.00

Winston is a 54-year-old male with a history of treatment-resistant schizophrenia. His care coordinator informs you that his most recent clozapine plasma level after repeat was 500 ng/mL. There are no signs of relapse and he appears to be in good physical health with no signs of toxicity. What could be your course of action?

Select one:

- ☐ Continue clozapine with anticonvulsant prophylaxis
- ☐ Psychosocial intervention
- ☐ Stop clozapine
- ☐ Reduce clozapine and add another antipsychotic medication
- ☐ Reduce clozapine and consider anticonvulsant prophylaxis

Your answer is incorrect.

Clozapine plasma level in the range of 250 to 350 ng/mL is a reasonable target for patients with schizophrenia. Reduced seizure threshold is a risk of high clozapine plasma levels. The advice in this case would be for cautious dose reduction of clozapine. Anticonvulsant prophylaxis (sodium valproate) should also be considered. Carbamazepine and phenytoin can increase the risk of agranulocytosis; concomitant use with clozapine is preferably avoided. Ref: De Berardis et al. Therapeutic advances in drug safety. 2018 May;9(5):237-56.

The correct answer is: Reduce clozapine and consider anticonvulsant prophylaxis

Question **42**

Not answered

Marked out of 1.00

Maria is a 33-year-old with a newly diagnosed treatment-resistant schizophrenia. You review her in your outpatient clinic and are considering commencing her on clozapine. You review her past medical history. Which of the following medical conditions is a contraindication for clozapine therapy?

Select one:

- ☐ Paralytic ileus
- ☐ Diabetes Mellitus
- ☐ Epilepsy
- ☐ Asthma
- ☐ Rheumatoid arthritis

Your answer is incorrect.

Clozapine therapy is associated with gastrointestinal hypomotility, resulting in constipation (very common) to the potentially fatal paralytic ileus (very rare). Paralytic ileus is an obstruction of the intestine due to paralysis of the intestinal muscles. (Ref: Cohen D. Clozapine and gastrointestinal hypomotility. CNS drugs. 2017 Dec 1;31(12):1083-91.)

The correct answer is: Paralytic ileus

Question **43**

Not answered

Marked out of 1.00

Carlos is a 46-year-old inpatient with a history of bipolar affective disorder. He presents as severely depressed. What is the best medication for him?

Select one:

- ☐ Amitriptyline
- ☐ Risperidone
- ☐ Fluoxetine with olanzapine
- ☐ Duloxetine
- ☐ Sodium valproate

Your answer is incorrect.

If a person with bipolar affective disorder develops moderate or severe bipolar depression and is not taking a drug to treat their bipolar disorder, offer fluoxetine combined with olanzapine, or quetiapine on its own, depending on the person's preference and previous response to treatment (Ref: <https://www.nice.org.uk/guidance/cg185/>).

The correct answer is: Fluoxetine with olanzapine

Question **44**

Not answered

Marked out of 1.00

Lucianna is a 36-year-old female with a history of paranoid schizophrenia. She is 36-weeks-pregnant. When reviewing her in your perinatal mental health outpatient clinic, you are concerned that she is presenting with symptoms of a psychotic relapse. She is currently not taking any psychotropic medications. Which of the following would be the most appropriate treatment?

Select one:

- ☐ Risperidone
- ☐ Aripiprazole
- ☐ Clozapine
- ☐ Flupenthixol
- ☐ Olanzapine

Your answer is incorrect.

The risks associated with untreated schizophrenia are high. It is generally advised that antipsychotics are used at every stage of pregnancy. Patients with a history of psychosis, who are maintained on an antipsychotic, particularly if they have frequent relapses, are best maintained on antipsychotics during and after their pregnancy. The most experience is with chlorpromazine, trifluoperazine, haloperidol, olanzapine and clozapine. Olanzapine is widely used by perinatal services in the UK.

The correct answer is: Olanzapine

Question 45

Not answered

Marked out of 1.00

In your community clinic, you review Reagan, a 27-year-old man with a history of paranoid schizophrenia. Last year following a relapse of his mental illness, he was admitted to an inpatient mental health unit. Since his discharge, he has engaged well with services and remains compliant with his antipsychotic medication (aripiprazole 15mg once a day). He would like to know how soon he could drive again. What would you advise him?

Select one:

- ☐ One month following discharge
- ☐ He must have remained well and stable for 6 weeks
- ☐ He must have remained well and stable for 3 months
- ☐ Immediately after discharge
- ☐ He won't be able to return to driving given his mental illness

Your answer is incorrect.

Patients who have experienced an acute psychotic episode must not drive during the acute phase of their illness and must notify the DVLA. Licensing may be considered if all of the following conditions are met; the patient: has remained well and stable for at least three months, adheres to any agreed treatment plan, has regained insight, is free from any medication effects that would impair driving, is subject to a suitable specialist report being favourable. Drivers with a history of instability and / or poor engagement will be required not to drive for a longer period before any relicensing (Info from gov.uk website).

The correct answer is: He must have remained well and stable for 3 months

Question 46

Not answered

Marked out of 1.00

Mr Jason Hughes is a 42-year-old man with a diagnosis of Down's Syndrome and mild intellectual disability. He largely lives independently with some community support and works for a local charity. He and his family feel it is important that he has a social role and life circumstances that are not too different from others of his age. Which of the following best describes this social phenomenon?

Select one:

- ☐ Social role valorisation
- ☐ Acclimatisation
- ☐ Deinstitutionalisation
- ☐ Assimilation
- ☐ Normalisation

Your answer is incorrect.

Normalisation refers to the principle that people with intellectual disabilities should be able to experience 'normal patterns' of everyday life, such as places of accommodation, and the range of day-to-day activities. Rather than forcing individuals to conform to societal norms, the principle argues for changing the environment, not the individual, to provide the required normalcy. Ref: Simpson MK. Power, ideology and structure: the legacy of normalization for intellectual disability. Social Inclusion. 2018;6(2):12-21.

The correct answer is: Normalisation

Question **47**

Not answered

Marked out of 1.00

You are working at a Learning Disabilities inpatient unit. You receive a referral for a new patient, 2- year-old Lizzy, who presents with repetitive hand movements that mimic hand-washing. Her parents tell you that she has no interest in playing, something they noticed in early infancy. She prefers to be on her own rather than the company of other children. Which of the following syndromes do you suspect Lizzy has?

Select one:

- ☐ Rett Syndrome
- ☐ Lesch Nyhan syndrome
- ☐ Angelman syndrome
- ☐ Prader Willi syndrome
- ☐ Smith Magenis syndrome

Your answer is incorrect.

Rett Syndrome is a rare genetic disorder affecting brain development. It results in severe mental and physical disability. Behavioural phenotypes include stereotypical hand movement described as hand-washing or hand- wringing and reduced interest in play in early infancy followed by autistic-like symptoms. Ref: Smeets EE, Pelc K, Dan B. Rett syndrome. Molecular syndromology. 2011;2(3-5):113-27.

The correct answer is: Rett Syndrome

Question 48

Not answered

Marked out of 1.00

Sophie Adams is a 5-year-old girl who has recently been diagnosed with a reactive attachment disorder after experiencing marked parental neglect during her early life. Which of the following factors is the most favourable outcome predictor in children with reactive attachment disorder?

Select one:

- ☐ Being resilient
- ☐ Doing well in school
- ☐ Being placed in an institution
- ☐ Providing an emotionally available attachment figure
- ☐ Having supportive peers

Your answer is incorrect.

Placement with a consistent set of carers is associated with better long-term outcomes, which is more likely with fostering than an institution. The most important intervention for young children diagnosed with RAD is ensuring that they are provided with an emotionally available attachment figure. Ref: Zeanah CH, et al., J Am Acad Child Adolesc Psychiatry. 2016 Nov;55(11):990-1003

The correct answer is: Providing an emotionally available attachment figure

Question 49

Not answered

Marked out of 1.00

Miss Emily Nicholson is a 30-year-old lady with Down's syndrome who has presented with gradual onset irritability, personality change and confusion. Her mother is concerned about her having developed Alzheimer's disease. Which of the following would be the most appropriate instrument to screen for dementia?

Select one:

- ☐ MOCA
- ☐ DMR
- ☐ ACE-R
- ☐ MMSE
- ☐ CAMCOG

Your answer is incorrect.

The Dementia Questionnaire for Mentally Retarded (DMR) would be the most appropriate for this scenario. The Dementia Screening Questionnaire for Individuals with Intellectual Disabilities (DSQIID) and Down Syndrome Dementia Scale (DSDS) would also be suitable. The Cambridge Cognition Examination (CAMCOG) is a neuropsychological battery and, while useful in this population, is not a screening tool. Ref: Elliott-King J, Shaw S, Bandelow S, Devshi R, Kassam S, Hogervorst E. A critical literature review of the effectiveness of various instruments in the diagnosis of dementia in adults with intellectual disabilities. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*. 2016 Jan 1;4:126-48.

The correct answer is: DMR

Question 50

Not answered

Marked out of 1.00

Which of the following genetic disorders are not explained by sporadic mutations?

Select one:

- ☐ Noonan syndrome
- ☐ Prader-Willi syndrome
- ☐ Williams syndrome
- ☐ Turner syndrome
- ☐ Rubinstein-Taybi syndrome

Your answer is incorrect.

Some medical conditions are caused by a single gene mutation passed onto a child by parents (e.g. autosomal recessive, autosomal dominant, X-linked). Some genetic mutations are known as sporadic (or de novo) mutations that are not inherited but occur during gametogenesis and postzygotically. All of the listed disorders, except Turner syndrome, are examples of sporadic mutations. Turner syndrome is not caused by a genetic mutation but by a missing or abnormal X chromosome (Ref: Acuna-Hidalgo R, et al. New insights into the generation and role of de novo mutations in health and disease. Genome biology. 2016 Dec;17(1):241.).

The correct answer is: Turner syndrome

Question **51**

Not answered

Marked out of 1.00

The MECP2 gene and protein appear to be essential for the normal function of nerve cells. Abnormalities within the gene are seen in which of the following disorders?

Select one:

- ☐ ADHD
- ☐ ASD
- ☐ Fragile X syndrome
- ☐ Rett Syndrome
- ☐ Angelman syndrome

Your answer is incorrect.

Rett syndrome is mainly found in girls with a prevalence of around 1/10,000. Around 6-18 months after birth, speech and motor function capabilities start to decrease. This is followed by seizures, growth retardation and cognitive and motor impairment. The MECP2 locus is X-linked and the disease-causing alleles are dominant. Ref: <https://ghr.nlm.nih.gov/gene/MECP2>

The correct answer is: Rett Syndrome

Question **52**

Not answered

Marked out of 1.00

Which of the following is the most common genetic disorder seen in patients with intellectual disability?

Select one:

- ☐ Noonan syndrome
- ☐ Rett syndrome
- ☐ Fragile X syndrome
- ☐ Turner syndrome
- ☐ Cornelia de Lange

Your answer is incorrect.

Of these, Fragile X is the most common genetic disorder causing intellectual disability. It occurs in around 1 in 4,000 males and 1 in 8,000 females. It is caused by a CGG triplet repeat within the Fragile X mental retardation 1 (FMR1) gene on the X chromosome.

The correct answer is: Fragile X syndrome

Question 53

Not answered

Marked out of 1.00

Richard Dawkins is a 6-year-old boy with a diagnosis of Autistic spectrum disorder (ASD) who is currently receiving input from his local CAMHS team. His parents want to know what the causes of ASD are. Which of the following would be the strongest predictor for developing ASD?

Select one:

- ☐ Microcephaly
- ☐ Poor parenting
- ☐ Maternal smoking during pregnancy
- ☐ Genetic predisposition
- ☐ Maternal obesity

Your answer is incorrect.

ASD has a very heritability, the proportion of symptoms in a population that is owing to genetic causes, with some estimates of this being up to 90%. Birth complications are associated with ASD, as are other factors such as maternal obesity, diabetes and C-section. Poor parenting and maternal smoking are not implicated.

The correct answer is: Genetic predisposition

Question **54**

Not answered

Marked out of 1.00

In your Learning Disabilities outpatient clinic you review Elliott, a 5-year-old boy with a profound learning disability. On examination, you notice that he has a slim face. He has a very warm and affectionate manner. His parents tell you that Elliott experiences anxiety. A report from his schoolteacher states that in class he is often hyperactive and lacks attention. There are concerns about his visuospatial and motor skills. Which of the following syndromes do you suspect Elliott has?

Select one:

- ☐ Prader Willi syndrome
- ☐ Smith Magenis syndrome
- ☐ Lesch Nyhan syndrome
- ☐ Williams syndrome
- ☐ Angelman syndrome

Your answer is incorrect.

Williams syndrome is a genetic disorder characterised by the following behavioural phenotypes: elfin-face, endearing, irrepresible and affectionate, anxiety disorder, phobias, inattention, hyperactivity, sleep disorder, hyperacusis and visuospatial / motor deficits.

The correct answer is: Williams syndrome

Question 55

Not answered

Marked out of 1.00

Which of the following genetic variants of Down syndrome shows less cognitive impairment and milder features than other variants?

Select one:

- ☐ Mosaicism
- ☐ Duplication of a portion of chromosome 21
- ☐ Full trisomy of chromosome 21
- ☐ Robertsonian translocation
- ☐ Both translocation and mosaicism

Your answer is incorrect.

Mosaicism (a mixture of normal and trisomic cell lines) is a variant of Down syndrome. Most individuals with Down syndrome have mild (IQ 50-69) or moderate (IQ 35-50) intellectual disability with some cases having severe (IQ 20-35) difficulties. Those with Mosaic Down syndrome typically have IQ scores that are 10-30 points higher (Ref: Megarbane A. et al., The intellectual disability of trisomy 21: differences in gene expression in a case series of patients with lower and higher IQ. European Journal of Human Genetics. 2013 Nov;21(11):1253.).

The correct answer is: Mosaicism

Question **56**

Not answered

Marked out of 1.00

Snags and dilemmas are present in which modality of psychotherapy?

Select one:

- ☐ CAT
- ☐ Family therapy
- ☐ CBT
- ☐ Psychodynamic psychotherapy
- ☐ Interpersonal therapy

Your answer is incorrect.

Snags are negative anticipations of the future consequences of actions that lead to halting of a procedure even before it is tested or run. Dilemmas are false choices or unreasonably narrowed options whereby the individual acts as though available action or possible roles were limited to polarised alternatives. Both are faulty procedural sequences, a theoretical base of cognitive analytic therapy (CAT). Ref: Leiman M. The development of cognitive analytic therapy. International Journal of Short-Term Psychotherapy. 1994 Apr;9(2/3):67-82.

The correct answer is: CAT

Question **57**

Not answered

Marked out of 1.00

Mr Peter Taylor is a 38-year-old man with a moderately severe intellectual disability who has been living in a residential placement for several years. He has been showing self-injurious behaviour and anger outbursts over the past 6 months. Which of the following will be most helpful in addressing this behaviour?

Select one:

- ☐ Functional behavioural analysis
- ☐ Risperidone
- ☐ Interpersonal therapy
- ☐ Cognitive therapy
- ☐ Behavioural therapy

Your answer is incorrect.

Functional behavioural assessment is a precise description of a given behaviour, its context, and its consequences, with the intent of understanding the factors influencing it. It is appropriate in the management of challenging behaviour in people with intellectual disability who may struggle to describe what drives some of their behaviours. Ref: Tassé MJ. Functional behavioural assessment in people with intellectual disabilities. *Current Opinion in Psychiatry*. 2006 Sep 1;19(5):475-80.

The correct answer is: Functional behavioural analysis

Question **58**

Not answered

Marked out of 1.00

Miss Baker is a 26-year-old woman who was made redundant due to issues around her work performance. She refuses to acknowledge that there is any issue with her work performance. She believes that she was sacked because one of her colleague was gossiping about her. After her dismissal, Miss Baker posted abusive messages against this colleague on her Facebook page. Which of the following is the defence mechanism being used?

Select one:

- ☐ Projective identification
- ☐ Reaction formation
- ☐ Denial
- ☐ Displacement
- ☐ Repression

Your answer is incorrect.

Displacement involves redirecting emotion to a safer outlet; the separation of emotion from its real object and redirection of the intense emotion toward someone or something that is less offensive or threatening in order to avoid dealing directly with what is frightening or threatening. In this case, the patient finds it threatening to accept that they may have been sacked for poor work performance but easier to blame another individual.

The correct answer is: Displacement

Question 59

Not answered

Marked out of 1.00

Mr Gregory Daniels is a 25-year-old man who is currently serving a 9-month sentence of ABH (Actual bodily harm). He has been admitted to the medical wing after cutting his wrist deeply, requiring suturing and dressing. Which of the following treatments would be most appropriate for him?

Select one:

- ☐ Interpersonal therapy
- ☐ Cognitive behavioural therapy
- ☐ Mentalisation based therapy
- ☐ Mindfulness training
- ☐ Dialectical behavioural therapy

Your answer is incorrect.

Dialectical behavioural therapy has the most evidence for managing self-harm and suicidal behaviours in a prison context. CBT has some evidence as well, as does improved staff training.

The correct answer is: Dialectical behavioural therapy

Question **60**

Not answered

Marked out of 1.00

Which of the following is a historical item on the HCR-20 scale?

Select one:

- ☐ Unstable relationships
- ☐ Active symptoms
- ☐ Non compliance
- ☐ Impulsivity
- ☐ Exposure to alcohol

Your answer is incorrect.

The Structured Clinical Risk assessment (HCR-20) is a popular structured clinical assessment tool for violence risk. HCR-20 (Historical, Clinical and Risk) shows good inter-rater reliability. It has 10 historical items, 5 clinical items and 5 risk management items. Historical items include previous violence history, young age at first incident, unstable relationships, major mental illness, substance use, psychopathy, employment issues, personality disorder, early maladjustment and previous supervision failure.

The correct answer is: Unstable relationships

Question **61**

Not answered

Marked out of 1.00

Diminished responsibility is a partial criminal defence for circumstances when an individual was not fully responsible for their actions due to mental disorder. Diminished responsibility is a defence for which of the following crimes?

Select one:

- ☐ Rape
- ☐ Damage to property
- ☐ Grievous bodily harm
- ☐ Shoplifting
- ☐ Murder

Your answer is incorrect.

In English law, diminished responsibility is one of the partial defenses that reduce the level of a homicidal act from murder to manslaughter if successful (termed "voluntary" manslaughter for these purposes). This allows the judge sentencing discretion, e.g. to impose a hospital order under S37 of the Mental Health Act to ensure treatment rather than a criminal punishment as in other cases.

The correct answer is: Murder

Question **62**

Not answered

Marked out of 1.00

Which of the following is NOT a clinical item on the HCR-20 scale?

Select one:

- ☐ Non compliance
- ☐ Negative attitudes to health services
- ☐ Active symptoms
- ☐ Impulsivity
- ☐ Treatment unresponsiveness

Your answer is incorrect.

The HCR-20 structured clinical assessment tool has been useful in predicting inpatient violence and community violence in discharged patients. Negative attitudes to health services, active symptoms, impulsivity, treatment unresponsiveness and lack of insight are all clinical items. Ref: Douglas KS, Guy LS, Reeves KA, Weir J. HCR-20 violence risk assessment scheme: Overview and annotated bibliography.

The correct answer is: Non compliance

Question 63

Not answered

Marked out of 1.00

Iris is an 83-year-old female with a diagnosis of Alzheimer's Disease. She was previously treated with donepezil but experienced side effects from this. Staff at the nursing home in which she resides complain that Iris has become increasingly agitated. You consider commencing memantine. Which of the following is a contraindication to memantine therapy?

Select one:

- ☐ Asthma
- ☐ Severe Alzheimer's disease
- ☐ eGFR < 5 mL/minute
- ☐ Epilepsy
- ☐ Gastroesophageal reflux disease

Your answer is incorrect.

Memantine is an NMDA antagonist licensed for moderate to severe dementia in Alzheimer's Disease. It is excreted through the kidneys and contraindicated in individuals with an eGFR < 5mL/minute (Ref: Moritoyo T, et al. Effect of renal impairment on the pharmacokinetics of memantine. Journal of pharmacological sciences. 2012;119(4):324-9).

The correct answer is: eGFR < 5 mL/minute

Question **64**

Not answered

Marked out of 1.00

Jean-Luc is a 65-year-old former teacher with a history of Parkinson's Disease. He presents to your outpatient clinic with delusional beliefs and auditory hallucinations. You diagnose him with late-onset schizophrenia and are keen to commence an antipsychotic medication. Which of the following would be the most appropriate antipsychotic medication to prescribe?

Select one:

- ☐ Risperidone
- ☐ Ziprasidone
- ☐ Quetiapine
- ☐ Olanzapine
- ☐ Haloperidol

Your answer is incorrect.

In patients with psychosis who have co-morbid Parkinson's disease, the prescription of antipsychotic medication can worsen symptoms of Parkinson's through dopamine blockade. Current advice states that atypical antipsychotics with a lower risk of EPSEs (e.g. quetiapine, clozapine) should be used, at lower doses (Chen J. Ment Health Clin. 2017 Nov; 7(6): 262–270.).

The correct answer is: Quetiapine

Question **65**

Not answered

Marked out of 1.00

You have reviewed Michael, a 71-year-old retired academic physicist in your memory service. He has a diagnosis of Mild Cognitive Impairment. He would like to know what the likelihood of developing Alzheimer's disease in the next one year. What advice do you give him?

Select one:

- ☐ 16-25%
- ☐ 26-35%
- ☐ 46-55%
- ☐ 36-45%
- ☐ 5-15%

Your answer is incorrect.

Patients with "mild cognitive impairment" (MCI) have an increased risk of developing Alzheimer disease at a rate of 10% to 15% per year. Nearly 80% of these patients convert to AD after approximately 6 years of follow-up. Ref: Lopez OL. Mild cognitive impairment. Continuum: Lifelong Learning in Neurology. 2013 Apr;19(2 Dementia):411.

The correct answer is: 5-15%

Question **66**

Not answered

Marked out of 1.00

In your outpatient clinic, you are reviewing Abdul. He is a 67-year-old man who works at an off-license store. He has had to take time off work due to a number of troubling symptoms that started abruptly. He suffered a number of falls at work due to loss of balance. He also had difficulties reading the cash register as 'everything seemed blurred'. His family members have noticed that he appears somewhat muddled and confused. Based on Abdul's symptoms, which of the following is the most likely diagnosis?

Select one:

- ☐ Cerebellar degeneration
- ☐ Wernicke's encephalopathy
- ☐ Korsakoff's syndrome
- ☐ Alcoholic dementia
- ☐ Wilson's disease

Your answer is incorrect.

Wernicke's encephalopathy is an acute condition characterised by a well-known triad of ataxia, ophthalmoplegia and confusion. It is caused by thiamine deficiency usually related to alcohol abuse. Pathological lesions are found in the mammillary bodies, thalamic nuclei, and the walls of the third ventricle. Ref: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3371089/>

The correct answer is: Wernicke's encephalopathy

Question **67**

Not answered

Marked out of 1.00

In which of the following dementias is REM sleep behaviour disorder a core feature?

Select one:

- ☐ Alzheimer's disease
- ☐ Mixed Dementia
- ☐ Lewy body dementia
- ☐ Vascular Dementia
- ☐ Fronto-temporal Dementia

Your answer is incorrect.

In rapid eye movement (REM) sleep behaviour disorder, 'dream-enacting behaviours' are reported. The affected individual physically acts out vivid, unpleasant dreams with vocal sounds and sudden, violent, arm and leg movements during the REM sleep.. It is a core feature of Dementia with Lewy bodies (DLB), and can precede cognitive decline by several years. Ref: Irfan M, Cherukuri CM, Khawaja IS, Schenck CH. Sleep Disturbances in Neurodegenerative Disorders. Psychiatric Annals. 2018 Jun 12;48(6):296-302.

The correct answer is: Lewy body dementia

Question 68

Not answered

Marked out of 1.00

'Discounting', the practice of weighting future gains and losses less heavily than those that occur in the present, is a common practice in the economic evaluation of health and other goods. In which of the following will 'discounting' in health economics have the biggest impact?

Select one:

- ☐ Use of ECT in depressed patient
- ☐ Use of ECT in depressed patient
- ☐ Clozapine prescription in schizophrenia patient
- ☐ Sending a patient to nursing home
- ☐ Early psychosis intervention in patient with first episode psychosis

Your answer is incorrect.

The principle of 'discounting' has greatest application to health-care programs for which most of the costs are incurred at the present moment and health benefits occur in the far future. These include public health programs such as screening and pediatric vaccination. Of the options presented, EIS input would represent a significant initial cost with long-term benefits, while most of the others involve ongoing costs, which provide simultaneous benefits. Ref: Attema AE, et al. Discounting in economic evaluations. *Pharmacoeconomics*. 2018 Jul 1;36(7):745-58. & Severens JL, Milne RJ. Discounting health outcomes in economic evaluation: the ongoing debate. *Value in health*. 2004 Jul;7(4):397-401.

The correct answer is: Early psychosis intervention in patient with first episode psychosis

Question 69

Not answered

Marked out of 1.00

A recently published economic analysis showed that the incremental cost effectiveness ratio for maternal age screening in Down's syndrome was 27% higher than the average ratio. When making cost calculations this study incorporated all the costs and benefits regardless of who incurs or obtains them. Whose perspective has been taken into account in this study?

Select one:

- ☐ Service provider's perspective
- ☐ Policy maker's perspective
- ☐ Patients' perspective
- ☐ For-profit perspective
- ☐ Societal perspective

Your answer is incorrect.

An economic evaluation can be carried out on the basis of a variety of different cost and benefit perspectives. The most commonly taken perspective is that of the health service. If the welfare of the whole of society, not just on the individuals or organisations directly involved in provisions of health, is taken into account then the study is said to have a societal perspective. Ref: Byford S., & Raftery J BMJ. 1998 May 16; 316(7143): 1529–1530.

The correct answer is: Societal perspective

Question 70

Not answered

Marked out of 1.00

Due to the difficulty in recruiting a full-time nurse for the ECT clinic, your hospital decides to provide the nurse who works at pain clinic to attend the ECT service on 'secondment'. Select the term that best describes the costs incurred by patients due to the absence of a nurse at the pain clinic as a result of this arrangement.

Select one:

- ☐ Intangible cost
- ☐ Zero-sum cost
- ☐ Direct cost
- ☐ Opportunity cost
- ☐ Indirect cost

Your answer is incorrect.

The opportunity cost is defined as the benefits that must be foregone when making any resource allocation decision. It refers to the fact that expending resources on one health care issue inevitably reduces the investment that could be made elsewhere. The opportunity cost of employing an ECT nurse is the benefit we forego by being unable to support the pain clinic. Ref: <https://emj.bmj.com/content/19/3/198>.

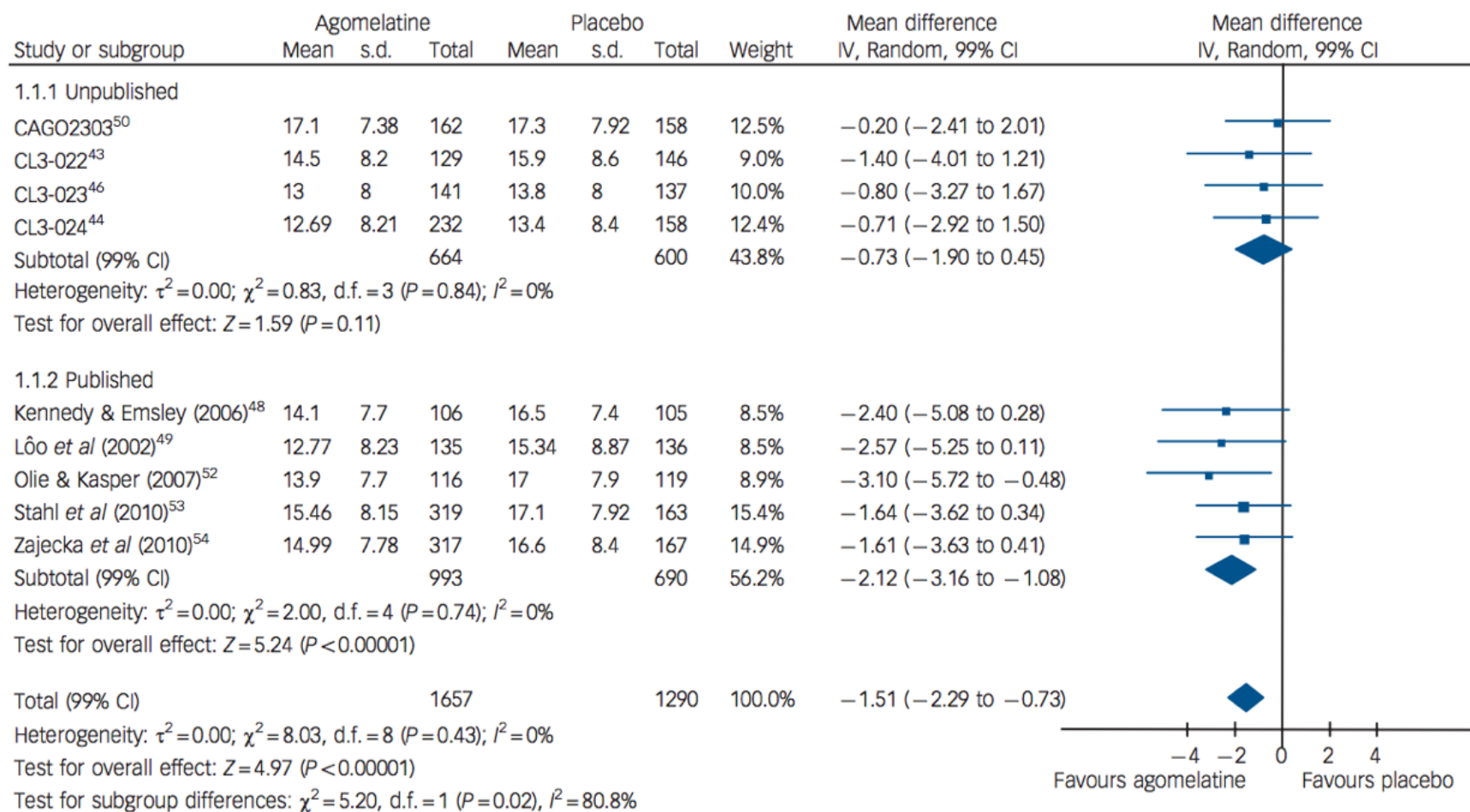
The correct answer is: Opportunity cost

Question **71**

Not answered

Marked out of 1.00

Randomised controlled trials comparing agomelatine with placebo in the treatment of unipolar major depression were systematically reviewed. Primary outcomes were (a) Hamilton Rating Scale for Depression (HRSD) score at the end of treatment (short-term studies) and (b) number of relapses (long-term studies). Meta-analyses included 10 acute-phase and 3 relapse prevention studies. Seven of the included studies were unpublished. Acute treatment with agomelatine was associated with a statistically significant superiority over placebo of -1.51 HRSD points (99% CI -2.29 to -0.73, nine studies). Data extracted from 3 relapse prevention studies failed to show significant effects of agomelatine over placebo (relative risk 0.78, 99% CI 0.41-1.48). Secondary efficacy analyses showed a significant advantage of agomelatine over placebo in terms of response (with no effect for remission). None of the negative trials were published and conflicting results between published and unpublished studies were observed. The attached figure summarises the results from acute phase studies included in this meta-analysis. Ref: Koesters M, et al. The British Journal of Psychiatry. 2013;203(3):179-87. Which of the following is the main objective of this study?



Select one:

- ☐ Reviewing published and unpublished evidence of the efficacy of agomelatine compared with placebo
- ☐ Indirectly estimating the efficacy of agomelatine as an adjunct to antidepressants
- ☐ Reviewing published and unpublished evidence of the cost effectiveness of agomelatine compared to other medications
- ☐ Reviewing published and unpublished evidence of the tolerability of agomelatine compared to placebo
- ☐ Reviewing published and unpublished evidence of the cost effectiveness of agomelatine compared with placebo

Your answer is incorrect.

The main objective of this study is to review published and unpublished evidence of the efficacy of agomelatine compared with placebo not other antidepressants. There is no cost-effectiveness data presented. Primary outcomes focus on efficacy not tolerability. The studies included here evaluate the use of agomelatine alone, not as an adjunct.

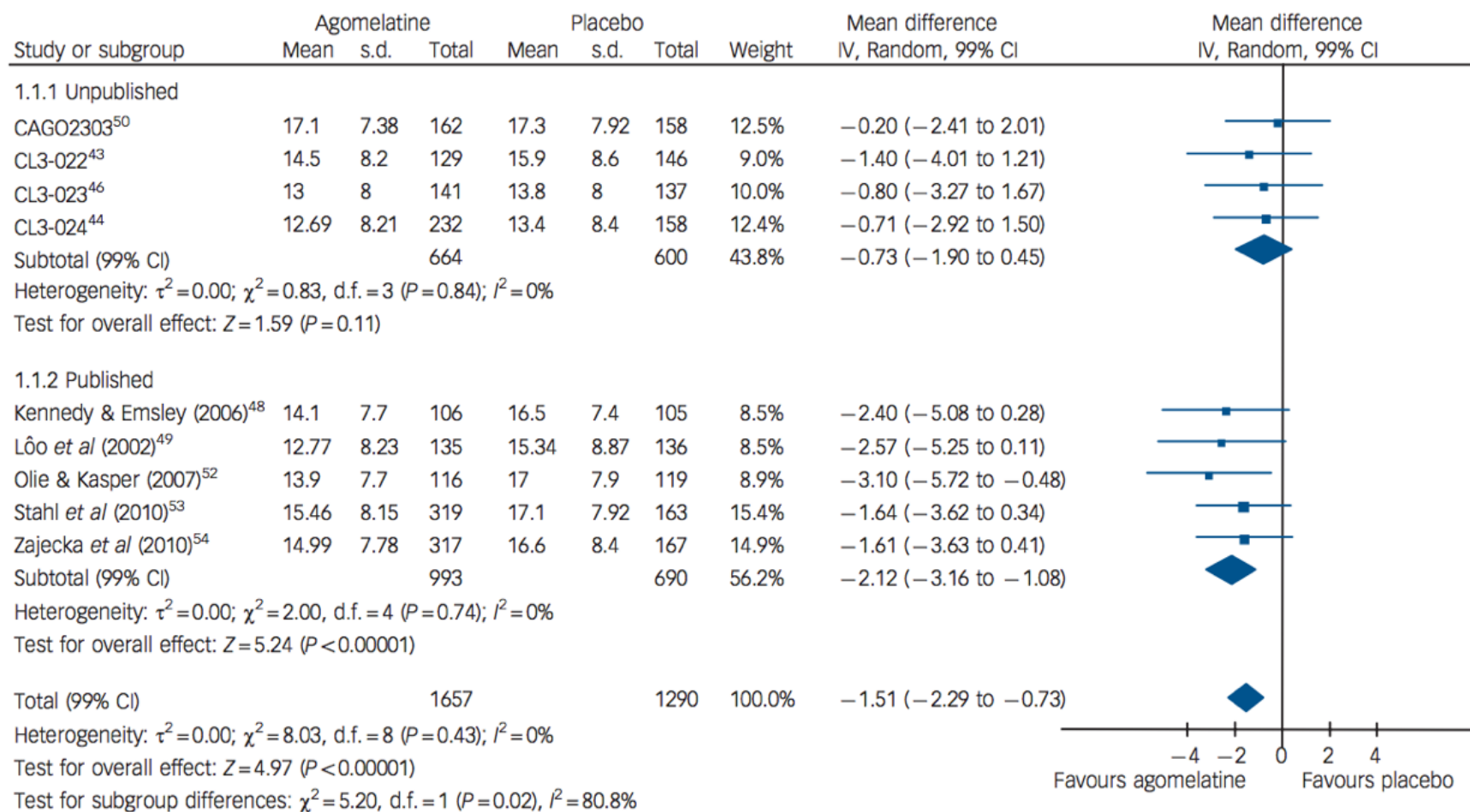
The correct answer is: Reviewing published and unpublished evidence of the efficacy of agomelatine compared with placebo

Question **72**

Not answered

Marked out of 1.00

Randomised controlled trials comparing agomelatine with placebo in the treatment of unipolar major depression were systematically reviewed. Primary outcomes were (a) Hamilton Rating Scale for Depression (HRSD) score at the end of treatment (short-term studies) and (b) number of relapses (long-term studies). Meta-analyses included 10 acute-phase and 3 relapse prevention studies. Seven of the included studies were unpublished. Acute treatment with agomelatine was associated with a statistically significant superiority over placebo of -1.51 HRSD points (99% CI -2.29 to -0.73, nine studies). Data extracted from 3 relapse prevention studies failed to show significant effects of agomelatine over placebo (relative risk 0.78, 99% CI 0.41-1.48). Secondary efficacy analyses showed a significant advantage of agomelatine over placebo in terms of response (with no effect for remission). None of the negative trials were published and conflicting results between published and unpublished studies were observed. The attached figure summarises the results from acute phase studies included in this meta-analysis. Ref: Koesters M, et al. The British Journal of Psychiatry. 2013;203(3):179-87. Which of the following conclusions is appropriate on the basis of this study?



Select one:

- ☐ Agomelatine is best used as an adjunct given the small-pooled effect
- ☐ Agomelatine is statistically superior to placebo in the treatment of depressive symptoms
- ☐ Agomelatine has a clinically important effect as an antidepressant
- ☐ Unpublished studies report larger effect sizes in favour of agomelatine than the published studies
- ☐ Agomelatine is comparatively less effective than other antidepressants when prescribed alone

Your answer is incorrect.

The main conclusion of this study is that there is a statistically significant effect for agomelatine in reducing symptoms of depression, when compared to placebo, but this effect appears to be too small to be clinically significant (-1.51 points gain in HRSD). Unpublished studies report smaller effect sizes that are not in favour of agomelatine, when compared to the published studies. But we cannot make any conclusions with respect to the use of agomelatine as an adjunct, or its effect in comparison with other antidepressants.

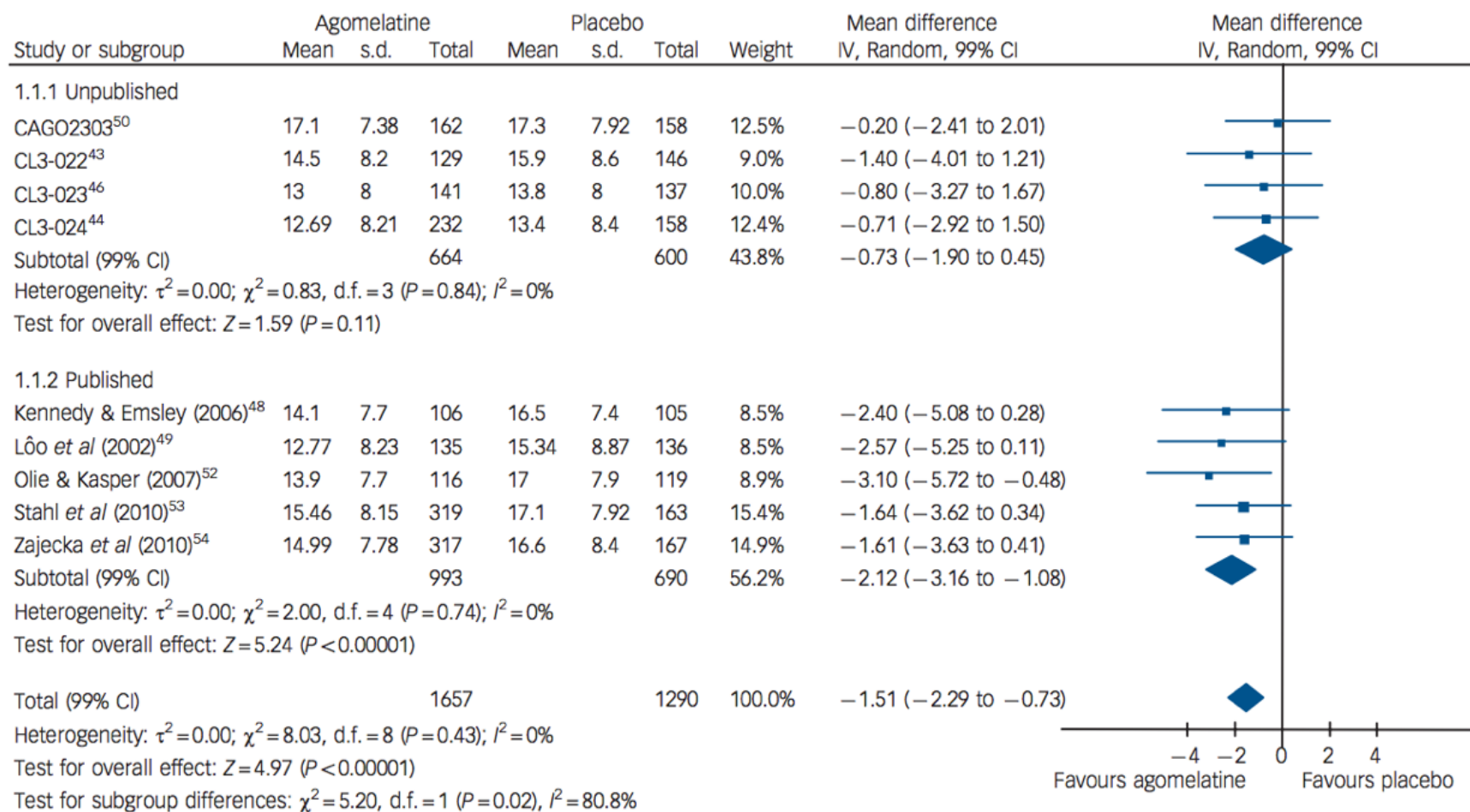
The correct answer is: Agomelatine is statistically superior to placebo in the treatment of depressive symptoms

Question **73**

Not answered

Marked out of 1.00

Randomised controlled trials comparing agomelatine with placebo in the treatment of unipolar major depression were systematically reviewed. Primary outcomes were (a) Hamilton Rating Scale for Depression (HRSD) score at the end of treatment (short-term studies) and (b) number of relapses (long-term studies). Meta-analyses included 10 acute-phase and 3 relapse prevention studies. Seven of the included studies were unpublished. Acute treatment with agomelatine was associated with a statistically significant superiority over placebo of -1.51 HRSD points (99% CI -2.29 to -0.73, nine studies). Data extracted from 3 relapse prevention studies failed to show significant effects of agomelatine over placebo (relative risk 0.78, 99% CI 0.41-1.48). Secondary efficacy analyses showed a significant advantage of agomelatine over placebo in terms of response (with no effect for remission). None of the negative trials were published and conflicting results between published and unpublished studies were observed. The attached figure summarises the results from acute phase studies included in this meta-analysis. Ref: Koesters M, et al. The British Journal of Psychiatry. 2013;203(3):179-87. Which of the following is a specific strength of this meta-analysis?



Select one:

- ☐ Equal weighting being given to all studies
- ☐ The inclusion of placebo-controlled trials
- ☐ Reporting random effects analysis
- ☐ The inclusion of unpublished studies
- ☐ Lack of differences between the 2 subgroups of studies

Your answer is incorrect.

The main strength of this study is the inclusion of unpublished studies, as this negates the effect of publication bias that is prevalent among meta-analyses. Unpublished studies report smaller effect sizes that are not in favour of agomelatine, when compared to the published studies. The studies are differentially weighted, and the reporting of random effects analysis is a more conservative approach in the absence of notable heterogeneity within subgroups.

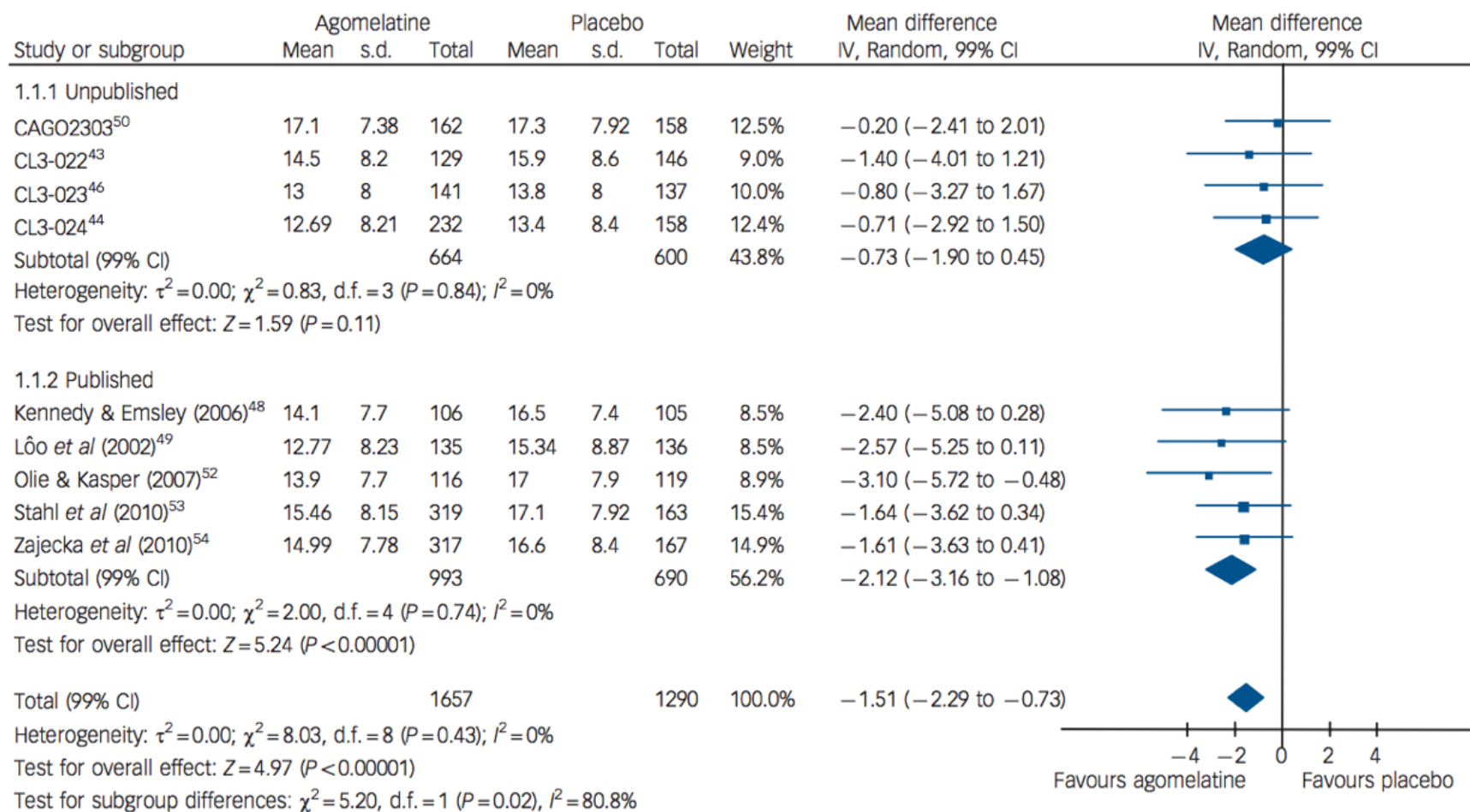
The correct answer is: The inclusion of unpublished studies

Question **74**

Not answered

Marked out of 1.00

Randomised controlled trials comparing agomelatine with placebo in the treatment of unipolar major depression were systematically reviewed. Primary outcomes were (a) Hamilton Rating Scale for Depression (HRSD) score at the end of treatment (short-term studies) and (b) number of relapses (long-term studies). Meta-analyses included 10 acute-phase and 3 relapse prevention studies. Seven of the included studies were unpublished. Acute treatment with agomelatine was associated with a statistically significant superiority over placebo of -1.51 HRSD points (99% CI -2.29 to -0.73, nine studies). Data extracted from 3 relapse prevention studies failed to show significant effects of agomelatine over placebo (relative risk 0.78, 99% CI 0.41-1.48). Secondary efficacy analyses showed a significant advantage of agomelatine over placebo in terms of response (with no effect for remission). None of the negative trials were published and conflicting results between published and unpublished studies were observed. The attached figure summarises the results from acute phase studies included in this meta-analysis. Ref: Koesters M, et al. The British Journal of Psychiatry. 2013;203(3):179-87. The authors of this meta-analysis could not find published negative trials reporting the lack of efficacy of agomelatine in relapse prevention. Which of the following explains this observation?



Select one:

- ☐ Publication bias in trials of agomelatine
- ☐ Robust efficacy of the drug agomelatine
- ☐ The common use of agomelatine as an adjunct antidepressant
- ☐ The lack of cost-effectiveness of agomelatine
- ☐ Lack of placebo effect in depression

Your answer is incorrect.

A major bias affecting meta-analyses is the bias against publishing trials with negative results that do not differentiate an active intervention vs. placebo. It is likely that negative results often lack sufficient power to detect a difference, thus mostly are of small sample size. Nevertheless, pooling them together will improve the power to detect a signal, if present, through a meta-analysis.

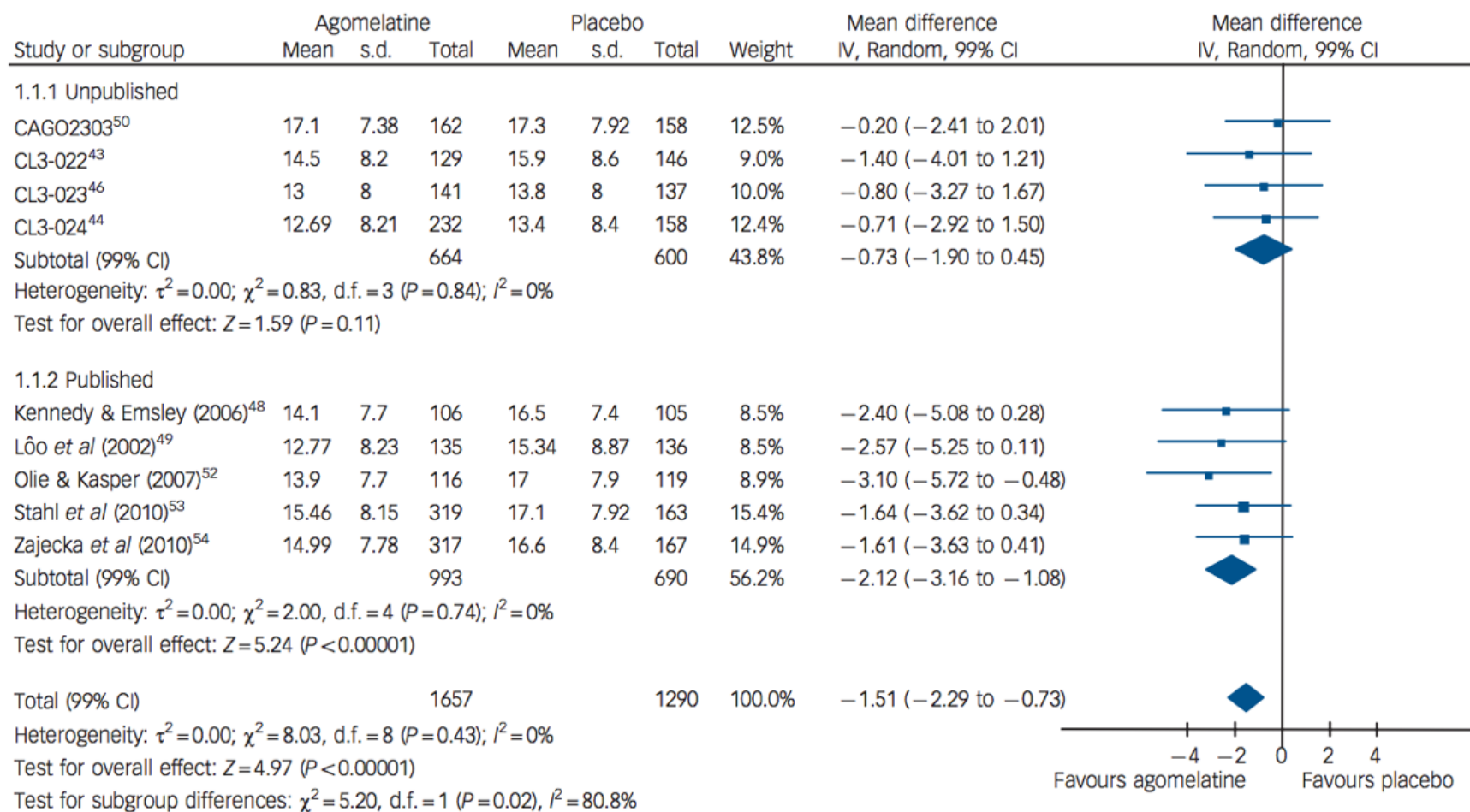
The correct answer is: The common use of agomelatine as an adjunct antidepressant

Question **75**

Not answered

Marked out of 1.00

Randomised controlled trials comparing agomelatine with placebo in the treatment of unipolar major depression were systematically reviewed. Primary outcomes were (a) Hamilton Rating Scale for Depression (HRSD) score at the end of treatment (short-term studies) and (b) number of relapses (long-term studies). Meta-analyses included 10 acute-phase and 3 relapse prevention studies. Seven of the included studies were unpublished. Acute treatment with agomelatine was associated with a statistically significant superiority over placebo of -1.51 HRSD points (99% CI -2.29 to -0.73, nine studies). Data extracted from 3 relapse prevention studies failed to show significant effects of agomelatine over placebo (relative risk 0.78, 99% CI 0.41-1.48). Secondary efficacy analyses showed a significant advantage of agomelatine over placebo in terms of response (with no effect for remission). None of the negative trials were published and conflicting results between published and unpublished studies were observed. The attached figure summarises the results from acute phase studies included in this meta-analysis. Ref: Koesters M, et al. The British Journal of Psychiatry. 2013;203(3):179-87. What type of analysis has been reported here?



Select one:

- ☐ Quasi-experimental meta-analysis
- ☐ Individual patient data network meta-analysis
- ☐ Literature-based network meta-analysis
- ☐ Literature-based standard direct meta-analysis
- ☐ Individual patient data meta-analysis

Your answer is incorrect.

Meta-analytical studies synthesise aggregate study level data obtained from study publications and report estimated effect size and the associated uncertainty for the original comparisons made in individual studies (e.g. treatment X vs. placebo). This is called a literature based, direct or pairwise meta-analysis, the evaluated pair of interventions being treatment X and placebo. An alternative approach is meta-analysis of individual patient data, in which the raw individual level data for each study are obtained from the authors and used for synthesis. We can also conduct a meta-analysis by pooling treatments that differ across trials, for example studies that compare X vs. placebo, Y vs. placebo and X vs. Z. We can then utilise this indirect evidence to estimate the relative treatment effect (ie, the treatment contrast) of X versus Y or Z vs. placebo, even in the absence of direct X vs. Y and Z vs. placebo studies. Such comparisons are called network meta-analyses. Ref: Riley RD, Lambert PC, Abo-Zaid G. Meta-analysis of individual participant data: rationale, conduct, and reporting. Bmj. 2010 Feb 5;340:c221.

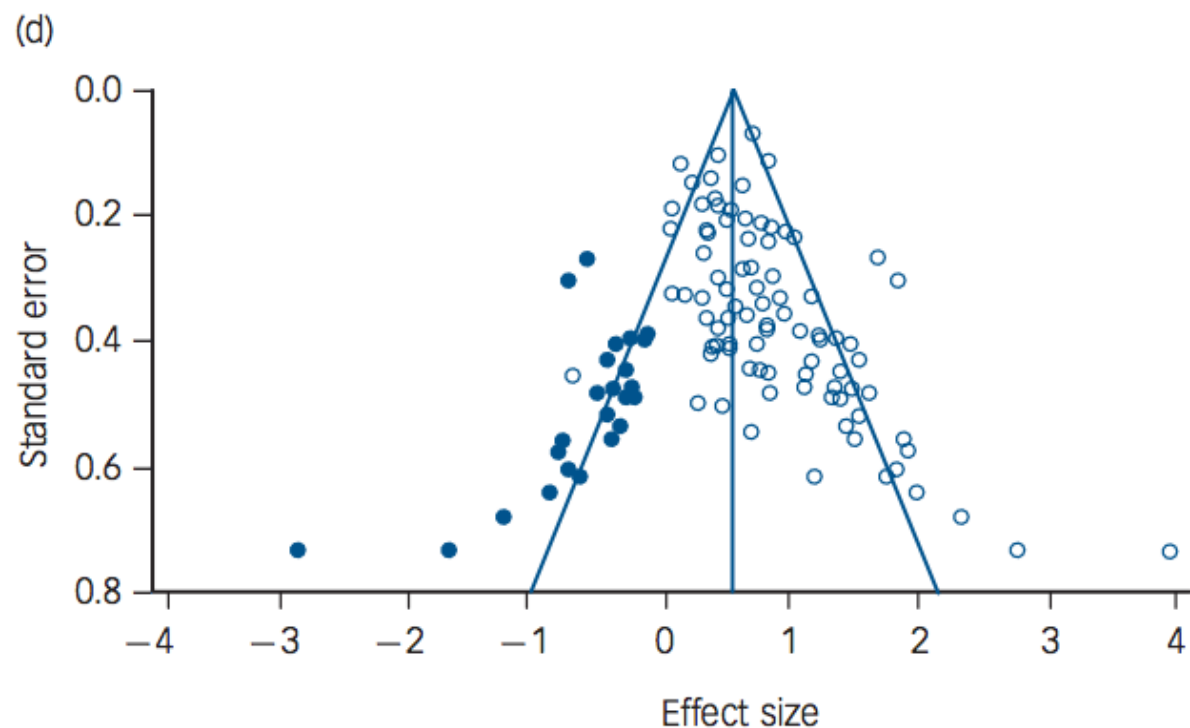
The correct answer is: Literature-based standard direct meta-analysis

Question 76

Not answered

Marked out of 1.00

In a meta-analysis published in 2010, authors examined effect sizes of 117 trials comparing the effect of psychotherapy and control conditions on depression. See the attached figure. The authors calculated adjusted effect sizes after imputing missing studies (shown as dark black circles, mostly on the left) while the trials included from systematic review are shown as white circles (on the right). What is the best term to describe the above plot?



Select one:

☐ Trim plot

- ☐ Forest plot
- ☐ Funnel plot
- ☐ Tray plot
- ☐ Fill plot

Your answer is incorrect.

The above figure is a Funnel plot. It helps in visualising the type of studies that are published in an area of inquiry.

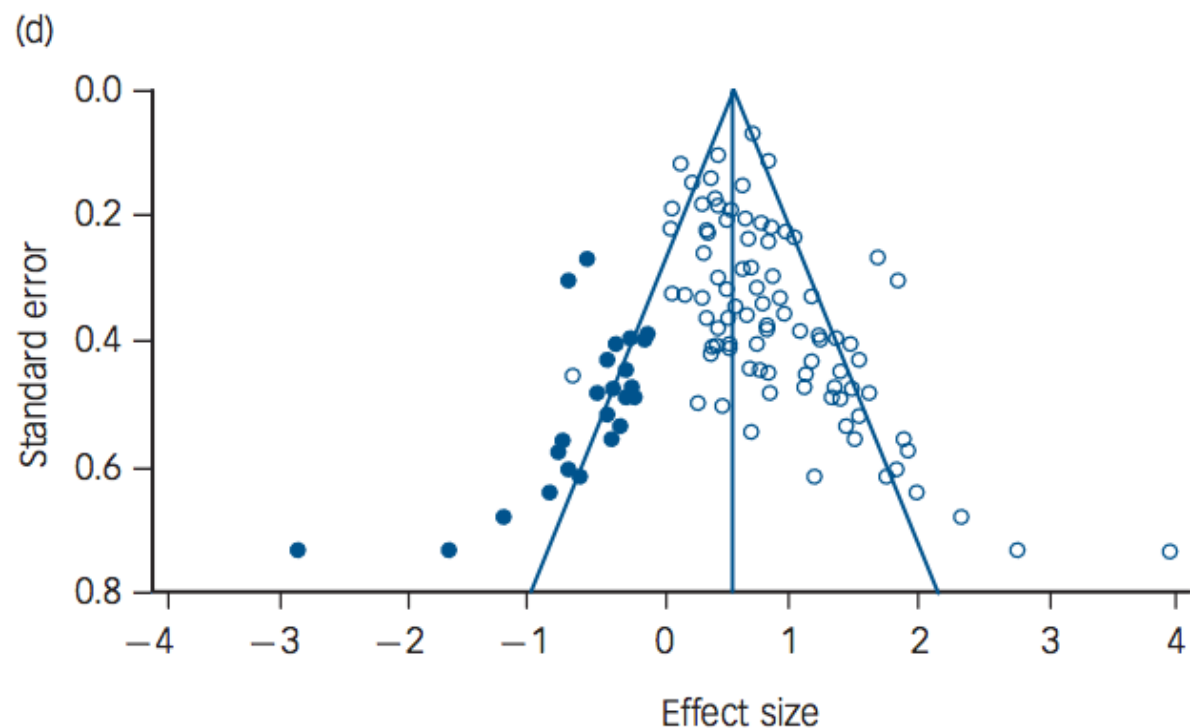
The correct answer is: Funnel plot

Question 77

Not answered

Marked out of 1.00

In a meta-analysis published in 2010, authors examined effect sizes of 117 trials comparing the effect of psychotherapy and control conditions on depression. See the attached figure. The authors calculated adjusted effect sizes after imputing missing studies (shown as dark black circles, mostly on the left) while the trials included from systematic review are shown as white circles (on the right). What type of bias causes the asymmetry in the distribution of the white circles?



Select one:

☐ Lead time bias

- ☐ Misclassification bias
- ☐ Publication bias
- ☐ Language bias
- ☐ Attrition bias

Your answer is incorrect.

Funnel plots are often used to examine for publication bias in the results of meta-analyses. Funnel plot asymmetry has a number of possible causes. In addition to reporting biases (publication bias, language or location bias), poor methodological quality that leads to spuriously inflated effects in smaller studies and true heterogeneity that induces a correlation between study sizes and intervention effects, can lead to funnel plot asymmetry. Ref: Sterne JA, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. BMJ. 2011 Jul 22;343:d4002.

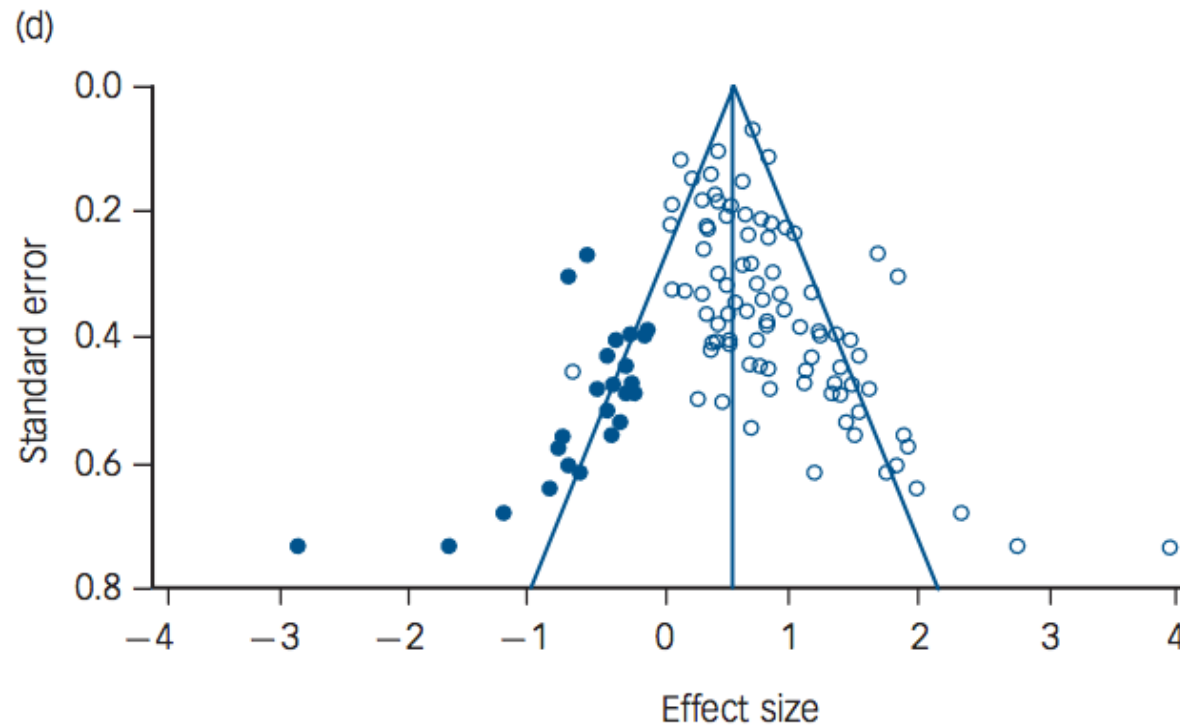
The correct answer is: Publication bias

Question 78

Not answered

Marked out of 1.00

In a meta-analysis published in 2010, authors examined effect sizes of 117 trials comparing the effect of psychotherapy and control conditions on depression. See the attached figure. The authors calculated adjusted effect sizes after imputing missing studies (shown as dark black circles, mostly on the left) while the trials included from systematic review are shown as white circles (on the right). What is the likely nature of the studies missing in this plot?



Select one:

☐ Studies with selection bias

- ☐ Studies with smaller sample size
- ☐ Studies that strongly favour psychotherapy
- ☐ Studies with longer follow-up duration
- ☐ Studies with larger sample size

Your answer is incorrect.

A major bias affecting meta-analyses is the bias against publishing trials with negative results that do not differentiate an active intervention vs. placebo (i.e. studies not favouring psychotherapy, in this example). It is likely that negative results often lack sufficient power to detect a difference, thus mostly are of small sample size. Individual study recruitment and follow-up duration are unlikely to produce this asymmetry. Ref: Sterne JA, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. BMJ. 2011 Jul 22;343:d4002.

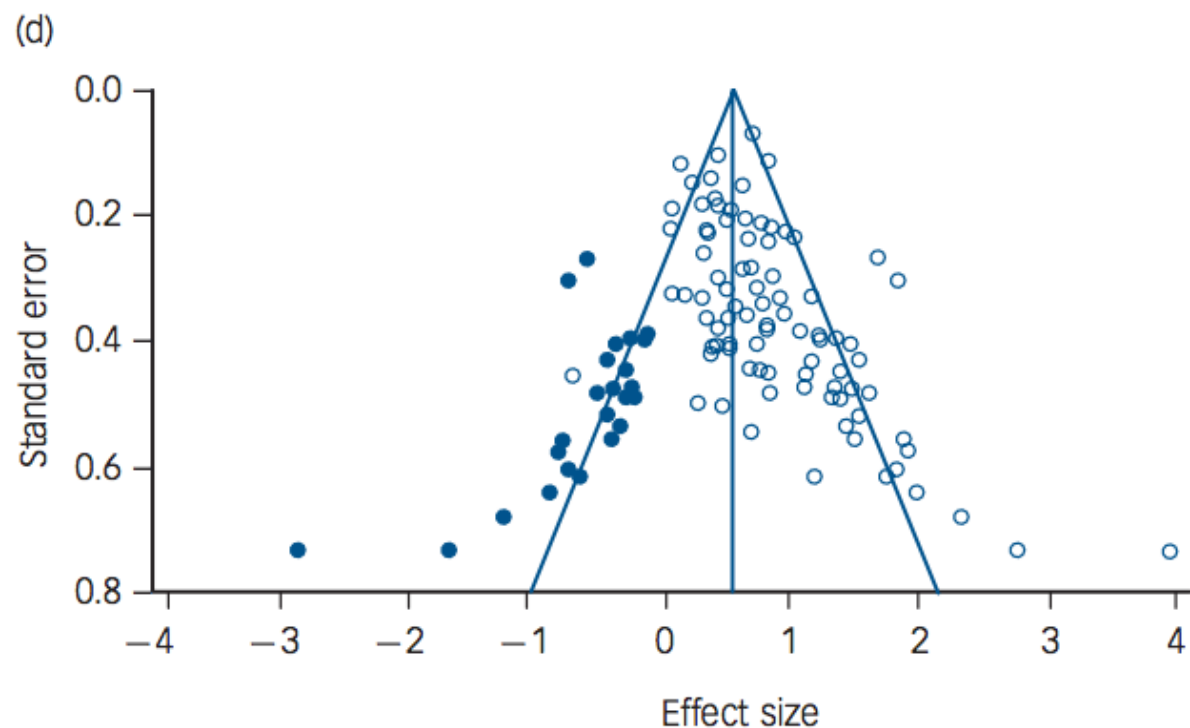
The correct answer is: Studies with smaller sample size

Question 79

Not answered

Marked out of 1.00

In a meta-analysis published in 2010, authors examined effect sizes of 117 trials comparing the effect of psychotherapy and control conditions on depression. See the attached figure. The authors calculated adjusted effect sizes after imputing missing studies (shown as dark black circles, mostly on the left) while the trials included from systematic review are shown as white circles (on the right). What is the likely effect of imputation on the pooled effect size reported from this meta-analysis?



Select one:

- ☐ The pooled result is more representative of the true effect

- ☐ The pooled result is not replicable any more
- ☐ The pooled result now underestimates the true effect
- ☐ The pooled result now overestimates the true effect
- ☐ Larger studies now influence the results more than smaller studies

Your answer is incorrect.

In a symmetric funnel plot indicating unbiased study selection for a meta-analysis, studies can be expected to be distributed symmetrically about the pooled effect size. In the presence of bias, it can be expected that the lower part of the plot will show a higher concentration of studies on one side of the mean than on the other. Replacing such missing studies by imputation procedures (e.g. Trim and Fill method) will drag the estimated effect size down from the high biased values to lower, truer and more representative values.

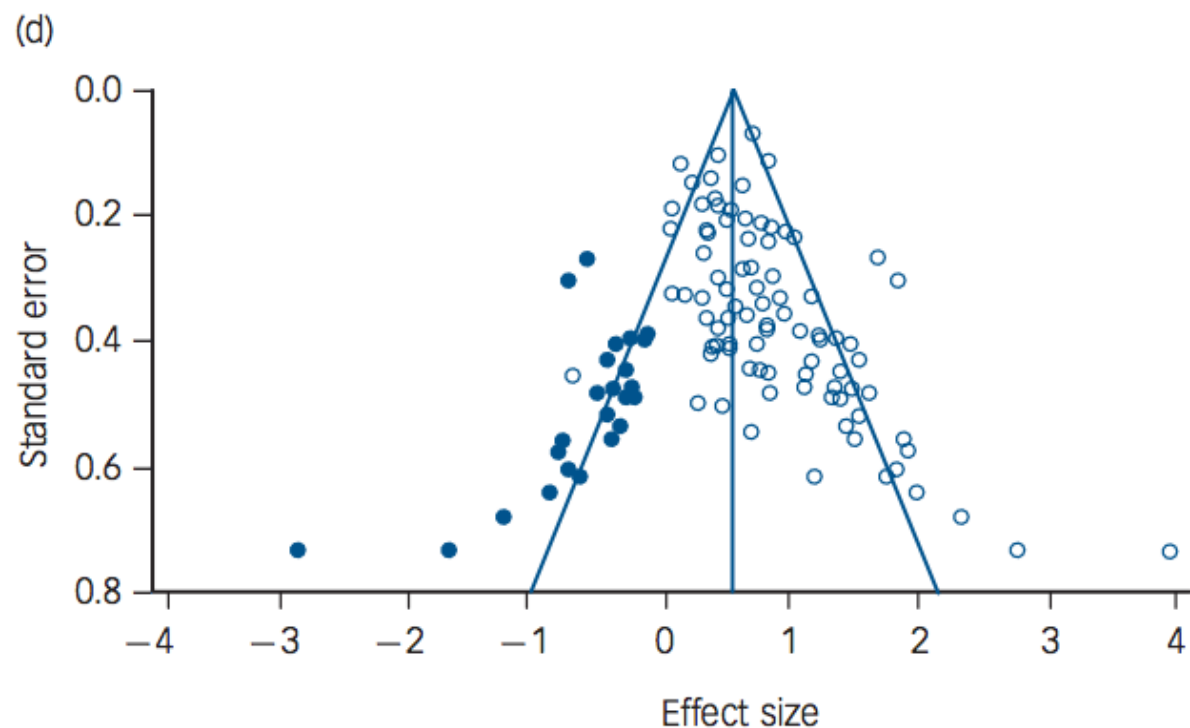
The correct answer is: The pooled result is more representative of the true effect

Question 80

Not answered

Marked out of 1.00

In a meta-analysis published in 2010, authors examined effect sizes of 117 trials comparing the effect of psychotherapy and control conditions on depression. See the attached figure. The authors calculated adjusted effect sizes after imputing missing studies (shown as dark black circles, mostly on the left) while the trials included from systematic review are shown as white circles (on the right). What is the assumption under which the imputation has been carried out?



Select one:

- ☐ Studies should be distributed close to the mean effect size

- ☐ Smaller studies are required to estimate the unbiased effect size
- ☐ Larger studies are required to estimate the unbiased effect size
- ☐ Studies with low standard error introduce result in biased effect size
- ☐ Studies should be equally distributed on both sides of the mean effect size

Your answer is incorrect.

The imputation procedure called the Duval & Tweedie trim and fill procedure, estimates the effect size after the publication bias has been taken into account (adjusted effect size), based on the assumption that the studies should be equally distributed on both sides of the mean effect size. Ref: Duval S, Tweedie R. Trim and fill: a simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. Biometrics. 2000 Jun;56(2):455-63.

The correct answer is: Studies should be equally distributed on both sides of the mean effect size

Question **81**

Not answered

Marked out of 1.00

Dopamine-replacement therapies in Parkinson's Disease (PD) are associated with increased incidence of impulse control disorders (ICDs). Three thousand ninety patients with treated PD receiving routine clinical care at 46 movement disorder centres were studied using the Massachusetts Gambling Screen score for current problem/pathological gambling, the Minnesota Impulsive Disorders Interview score for compulsive sexual behaviour and buying, and Diagnostic and Statistical Manual of Mental Disorders research criteria for binge-eating disorder. An ICD was identified in 13.6% of patients (gambling in 5.0%, compulsive sexual behaviour in 3.5%, compulsive buying in 5.7%, and binge-eating disorder in 4.3%), and 3.9% had 2 or more ICDs. Impulse control disorders were more common in patients treated with a dopamine agonist than in patients not taking a dopamine agonist (17.1% vs. 6.9%; odds ratio [OR], 2.72; 95% confidence interval [CI], 2.08-3.54; $P < .001$). Impulse control disorder frequency was similar for pramipexole and ropinirole (17.7% vs. 15.5%; OR, 1.22; 95% CI, 0.94-1.57; $P = .14$). Additional variables independently associated with ICDs were levodopa use, younger age, being unmarried, current cigarette smoking, and a family history of gambling problems. Dopamine agonist treatment in PD is associated with 2- to 3.5-fold increased odds of having an ICD. Ref: Weintraub et al., Arch Neurol. 2010;67(5):589-595. What type of study has been conducted?

Select one:

- ☐ Cohort study
- ☐ Cross-sectional survey
- ☐ Nested case-control study
- ☐ Controlled clinical trial
- ☐ Cluster randomised control trial

Your answer is incorrect.

3090 patients with treated PD receiving routine clinical care at several movement disorder centres were studied using standard questionnaires to ascertain the presence of ICD and treatment status and the association between the questionnaire results and the treatment status is reported. This is a multi-centre cross-sectional observation with no randomisation, experimental manipulation or prospective follow-up.

The correct answer is: Cross-sectional survey

Question 82

Not answered

Marked out of 1.00

Dopamine-replacement therapies in Parkinson's Disease (PD) are associated with increased incidence of impulse control disorders (ICDs). Three thousand ninety patients with treated PD receiving routine clinical care at 46 movement disorder centres were studied using the Massachusetts Gambling Screen score for current problem/pathological gambling, the Minnesota Impulsive Disorders Interview score for compulsive sexual behaviour and buying, and Diagnostic and Statistical Manual of Mental Disorders research criteria for binge-eating disorder. An ICD was identified in 13.6% of patients (gambling in 5.0%, compulsive sexual behaviour in 3.5%, compulsive buying in 5.7%, and binge-eating disorder in 4.3%), and 3.9% had 2 or more ICDs. Impulse control disorders were more common in patients treated with a dopamine agonist than in patients not taking a dopamine agonist (17.1% vs. 6.9%; odds ratio [OR], 2.72; 95% confidence interval [CI], 2.08-3.54; $P < .001$). Impulse control disorder frequency was similar for pramipexole and ropinirole (17.7% vs. 15.5%; OR, 1.22; 95% CI, 0.94-1.57; $P = .14$). Additional variables independently associated with ICDs were levodopa use, younger age, being unmarried, current cigarette smoking, and a family history of gambling problems. Dopamine agonist treatment in PD is associated with 2- to 3.5-fold increased odds of having an ICD. Ref: Weintraub et al., Arch Neurol. 2010;67(5):589-595. What can be concluded from this study?

Select one:

- ☐ Untreated PD is not associated with impulse control disorders
- ☐ A strong association between dopamine agonist use and impulse control disorders is present in patients with PD
- ☐ Dopamine agonists do not cause impulse control disorders in patients with PD
- ☐ Dopamine agonists cause impulse control disorders
- ☐ Dopamine agonists cause impulse control disorders in the presence of PD

Your answer is incorrect.

This is a multi-centre cross-sectional observation with no randomisation, experimental manipulation or prospective follow-up. As such, we cannot infer causality, but only the strength of association. A strong association between dopamine agonist use and impulse control disorders is present in patients with PD.

The correct answer is: A strong association between dopamine agonist use and impulse control disorders is present in patients with PD

Question 83

Not answered

Marked out of 1.00

Dopamine-replacement therapies in Parkinson's Disease (PD) are associated with increased incidence of impulse control disorders (ICDs). Three thousand ninety patients with treated PD receiving routine clinical care at 46 movement disorder centres were studied using the Massachusetts Gambling Screen score for current problem/pathological gambling, the Minnesota Impulsive Disorders Interview score for compulsive sexual behaviour and buying, and Diagnostic and Statistical Manual of Mental Disorders research criteria for binge-eating disorder. An ICD was identified in 13.6% of patients (gambling in 5.0%, compulsive sexual behaviour in 3.5%, compulsive buying in 5.7%, and binge-eating disorder in 4.3%), and 3.9% had 2 or more ICDs. Impulse control disorders were more common in patients treated with a dopamine agonist than in patients not taking a dopamine agonist (17.1% vs. 6.9%; odds ratio [OR], 2.72; 95% confidence interval [CI], 2.08-3.54; $P < .001$). Impulse control disorder frequency was similar for pramipexole and ropinirole (17.7% vs. 15.5%; OR, 1.22; 95% CI, 0.94-1.57; $P = .14$). Additional variables independently associated with ICDs were levodopa use, younger age, being unmarried, current cigarette smoking, and a family history of gambling problems. Dopamine agonist treatment in PD is associated with 2- to 3.5-fold increased odds of having an ICD. Ref: Weintraub et al., Arch Neurol. 2010;67(5):589-595. On the basis of this study, we cannot define the etiological risk factors of impulse control disorders in PD. This is because,

Select one:

- ☐ It is not a randomized study and confounders are not controlled for.
- ☐ It lacks sufficient power to establish etiological links.
- ☐ The data is heterogeneous, obtained from multiple centres.
- ☐ Biological investigations have not been conducted
- ☐ It is not a prospective study and temporal association cannot be made.

Your answer is incorrect.

As this cross-sectional observation does not have a prospective study design essential to infer temporal relationships between cause and effect, we cannot infer causality.

The correct answer is: It is not a prospective study and temporal association cannot be made.

Question 84

Not answered

Marked out of 1.00

Dopamine-replacement therapies in Parkinson's Disease (PD) are associated with increased incidence of impulse control disorders (ICDs). Three thousand ninety patients with treated PD receiving routine clinical care at 46 movement disorder centres were studied using the Massachusetts Gambling Screen score for current problem/pathological gambling, the Minnesota Impulsive Disorders Interview score for compulsive sexual behaviour and buying, and Diagnostic and Statistical Manual of Mental Disorders research criteria for binge-eating disorder. An ICD was identified in 13.6% of patients (gambling in 5.0%, compulsive sexual behaviour in 3.5%, compulsive buying in 5.7%, and binge-eating disorder in 4.3%), and 3.9% had 2 or more ICDs. Impulse control disorders were more common in patients treated with a dopamine agonist than in patients not taking a dopamine agonist (17.1% vs. 6.9%; odds ratio [OR], 2.72; 95% confidence interval [CI], 2.08-3.54; $P < .001$). Impulse control disorder frequency was similar for pramipexole and ropinirole (17.7% vs. 15.5%; OR, 1.22; 95% CI, 0.94-1.57; $P = .14$). Additional variables independently associated with ICDs were levodopa use, younger age, being unmarried, current cigarette smoking, and a family history of gambling problems. Dopamine agonist treatment in PD is associated with 2- to 3.5-fold increased odds of having an ICD. Ref: Weintraub et al., Arch Neurol. 2010;67(5):589-595. The authors conclude that ". Dopamine agonist treatment in PD is associated with 2- to 3.5-fold increased odds of having an ICD". The values 2 and 3.5 are obtained from

Select one:

- ☐ Confidence intervals of Odds Ratio values
- ☐ Range of Odds Ratio values for the different types of ICDs
- ☐ Adjusted and unadjusted Odds Ratio values
- ☐ Range of Odds Ratio values from individual movement disorder centres
- ☐ Confidence intervals of Relative Risk values

Your answer is incorrect.

The values 2 and 3.5 are approximations of the values from the 95% confidence interval limits in the statement - "Impulse control disorders were more common in patients treated with a dopamine agonist than in patients not taking a dopamine agonist (17.1% vs. 6.9%; odds ratio [OR], 2.72; 95% confidence interval [CI], 2.08-3.54; $P < .001$)".

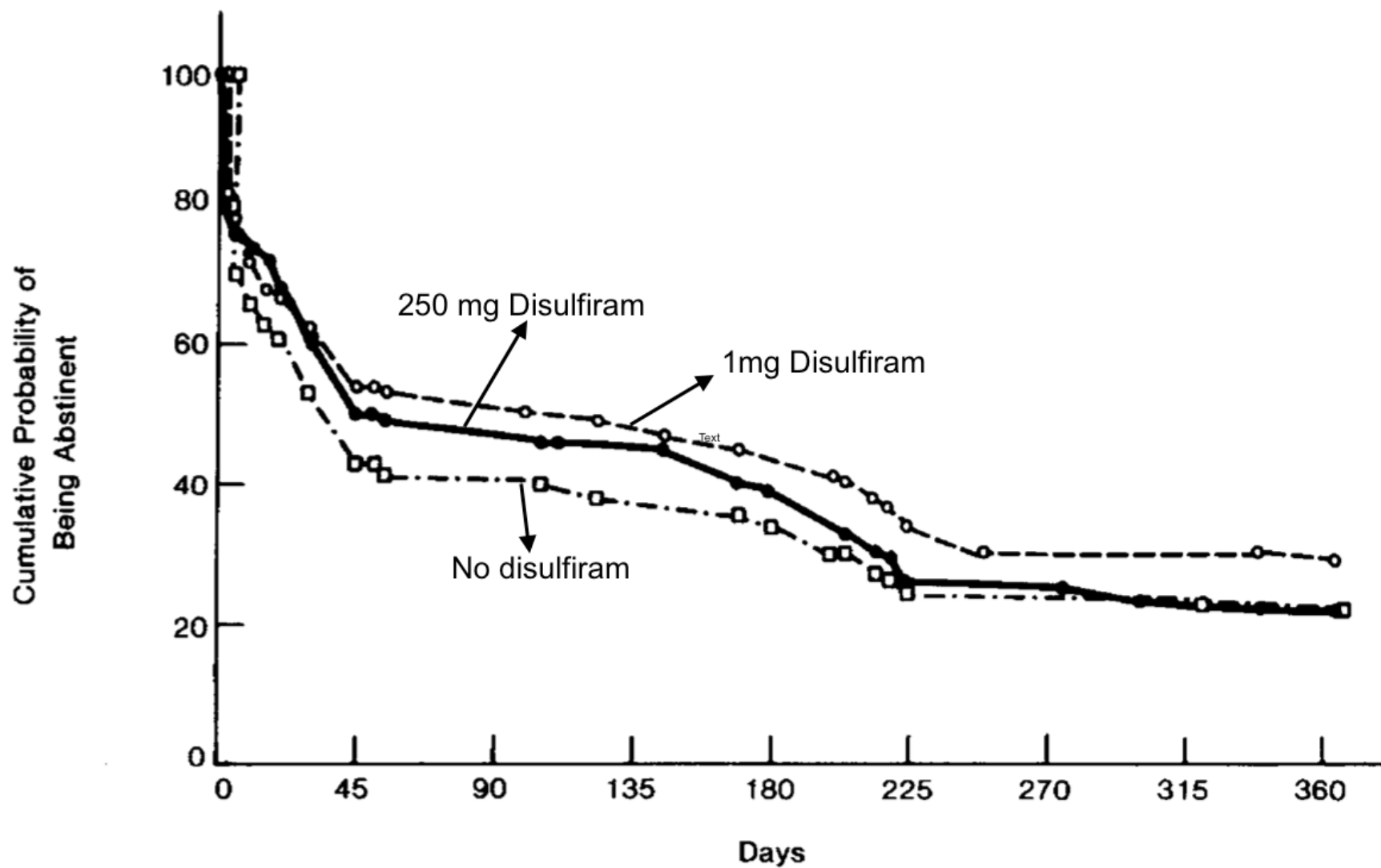
The correct answer is: Confidence intervals of Odds Ratio values

Question **85**

Not answered

Marked out of 1.00

The term that best describes the below graph is



Select one:

- ☐ Histogram
- ☐ Bar chart
- ☐ Box and dots
- ☐ Survival graph
- ☐ Pie graph

Your answer is incorrect.

A survival curve is a graph that shows the survival experience of a group of patients i.e. the proportion surviving an event of interest (here, the event to survive being a relapse to alcohol use).

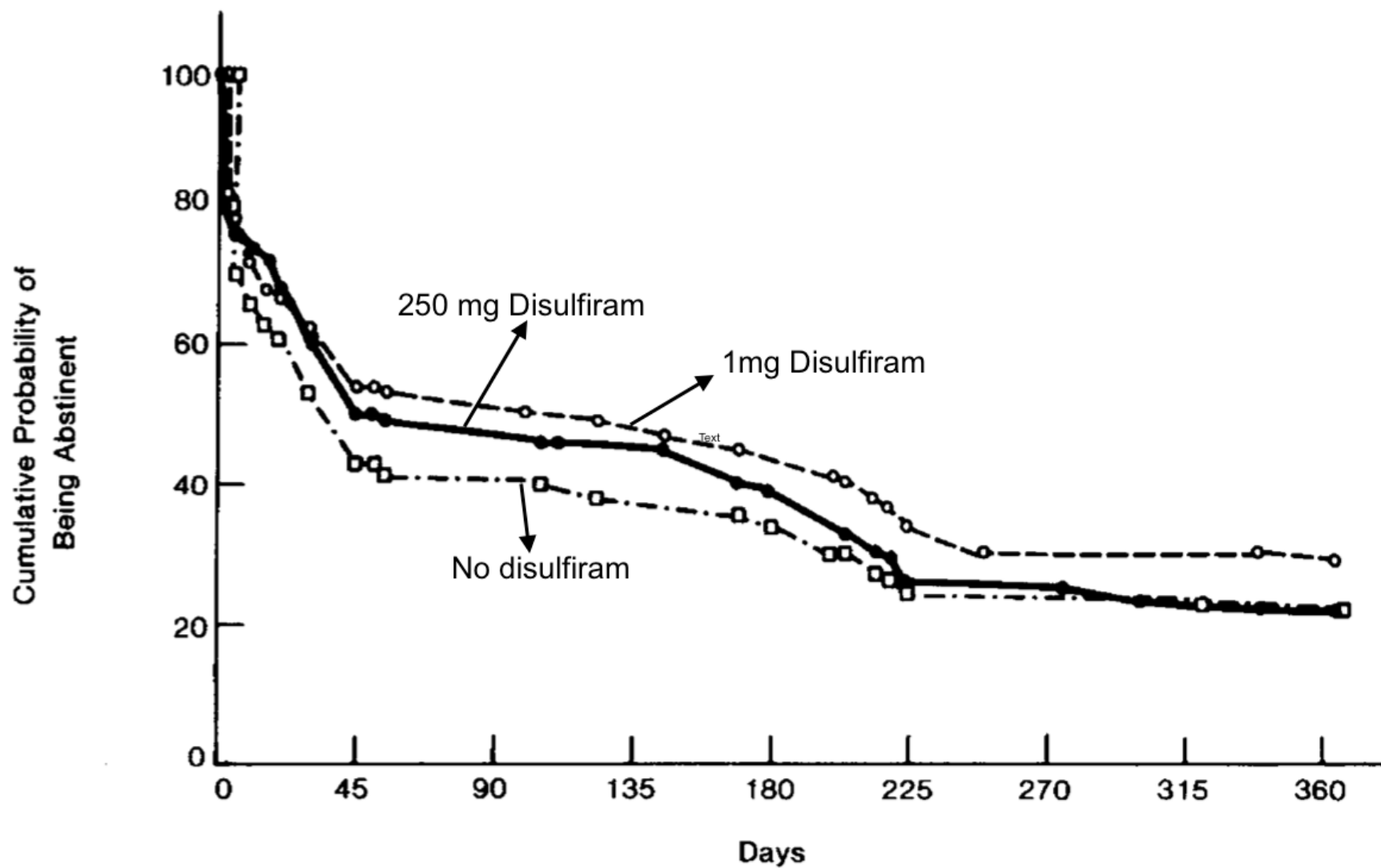
The correct answer is: Survival graph

Question **86**

Not answered

Marked out of 1.00

See the below graph. The probability for remaining abstinent up to 6 months after starting 250mg of disulfiram is nearly



Select one:

- ☐ 50%
- ☐ 40%
- ☐ 80%
- ☐ 20%
- ☐ 60%

Your answer is incorrect.

If we draw a straight vertical line at 180 days time point (=6 months), this line touches the 40% mark for the 250mg curve. So the probability of being abstinent at this time point was 40% for those who took 250mg of disulfiram.

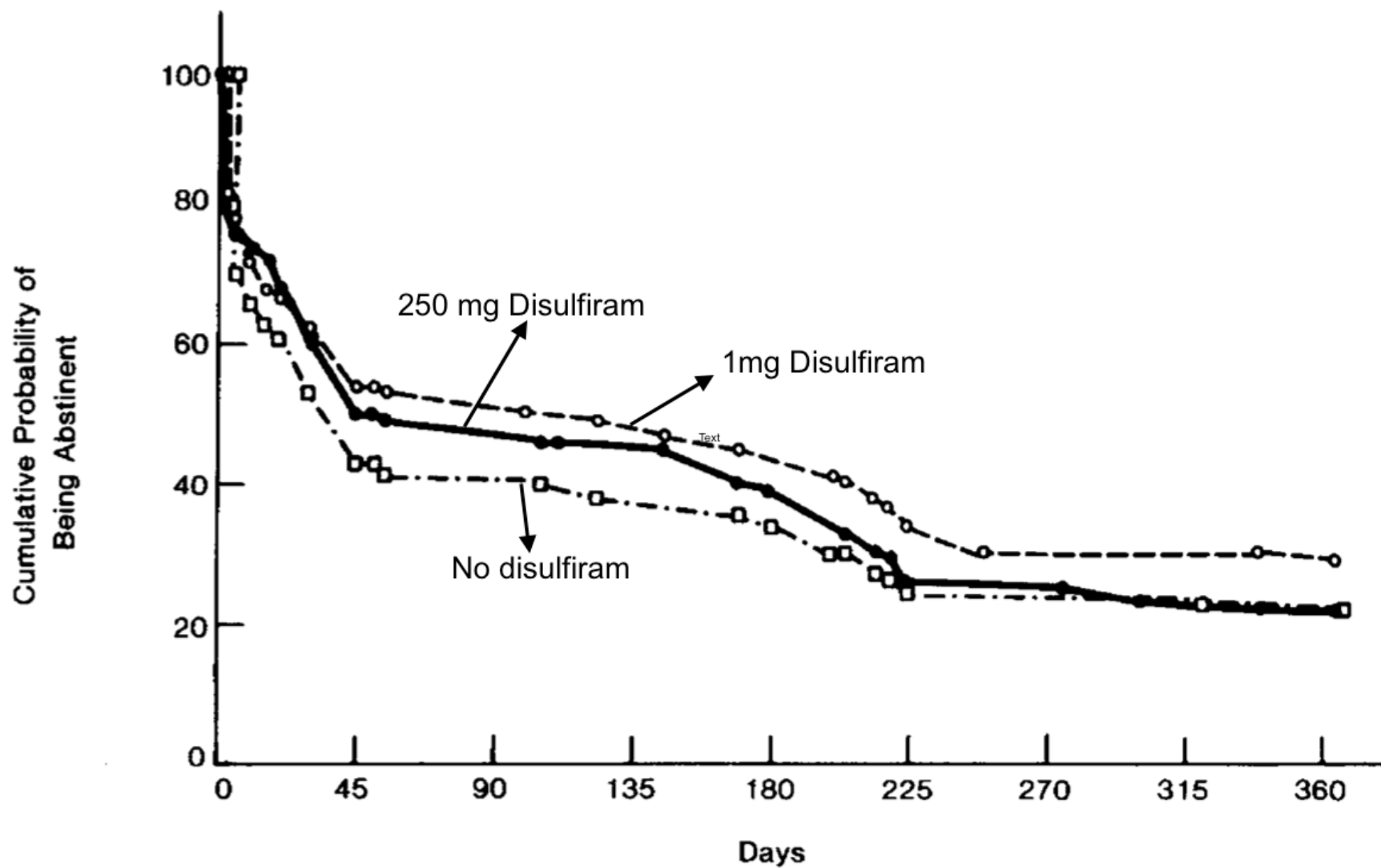
The correct answer is: 40%

Question **87**

Not answered

Marked out of 1.00

See the below graph. The probability for relapsing to alcoholism 1 year after starting 250mg of disulfiram is



Select one:

- ☐ Around 20%
- ☐ Nearly the same as the no disulfiram control group
- ☐ Nearly the same as the 1mg disulfiram group
- ☐ Significantly different from the control group
- ☐ Significantly different from the 1mg disulfiram group

Your answer is incorrect.

If we draw a straight vertical line at 360 days time point (=12 months approximately), this line touches the 20% mark for the 250mg curve. This value represents the probability of being abstinent; therefore, the probability of relapse is 100% minus 20%, i.e. nearly 80%. This value is nearly the same for the no disulfiram control group.

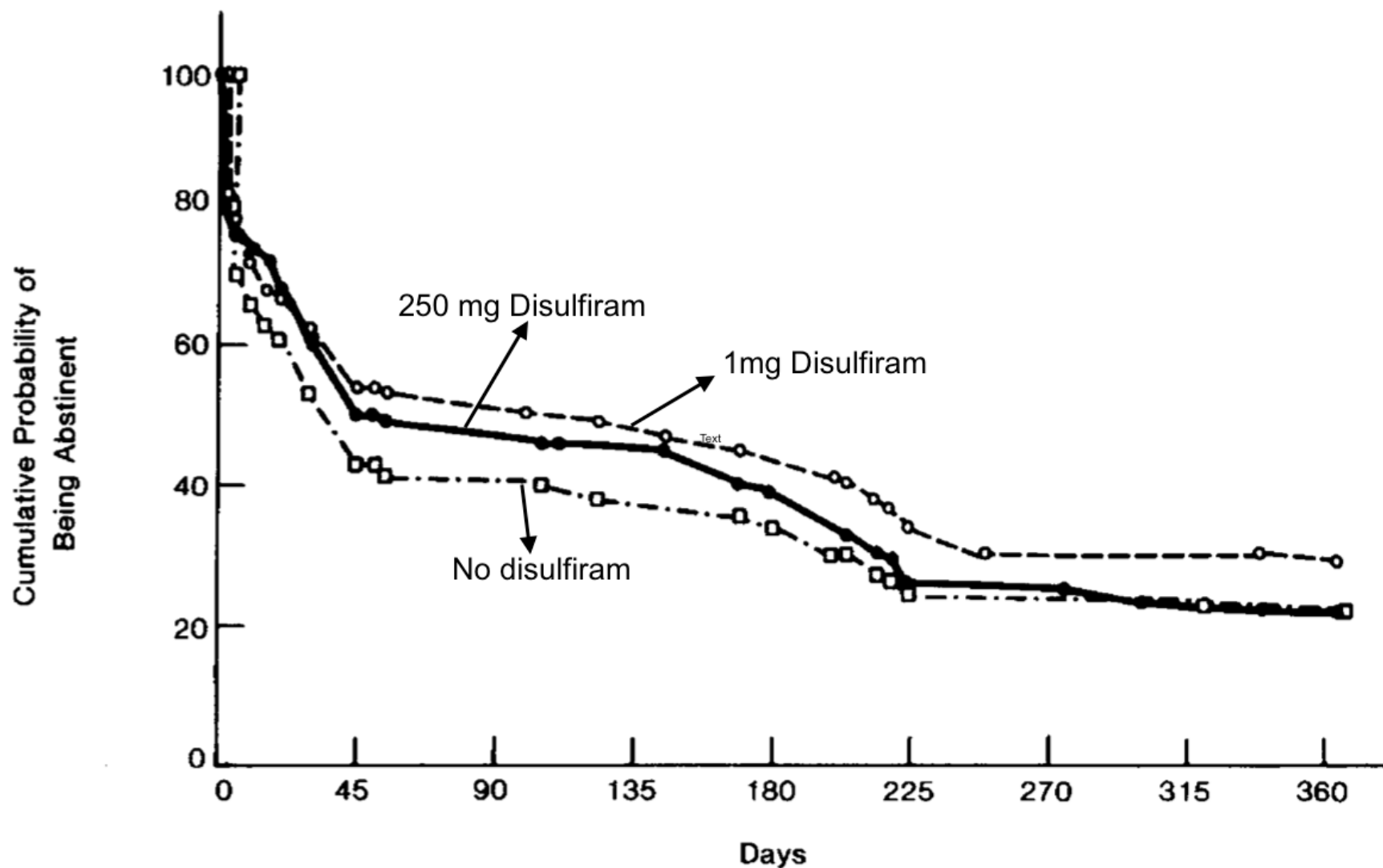
The correct answer is: Nearly the same as the no disulfiram control group

Question **88**

Not answered

Marked out of 1.00

See the below graph. The most reasonable conclusion that can be made from this study is



Select one:

- ☐ Disulfiram 250mg should be continued for more than 1 month to be beneficial
- ☐ Disulfiram 250mg is superior to placebo to promote abstinence only between 45th and 225th day of treatment
- ☐ Disulfiram 250mg reduces the probability of being abstinent from alcohol in the first month of use
- ☐ There are no beneficial effects for disulfiram 250mg during the 1 year of use.
- ☐ Disulfiram 250mg promotes abstinence from alcohol at least for 1 year

Your answer is incorrect.

The data shown here indicates that there are no statistically significant differences among the 3 groups in terms of time to relapse. Though visual inspection indicates some degree of separation among the groups between 45th and 225th days, this is not a significant difference as per the information provided in the text.

The correct answer is: There are no beneficial effects for disulfiram 250mg during the 1 year of use.

Question 89

Not answered

Marked out of 1.00

To explore the current practices and challenges in night-time care for people with dementia living in care homes in the UK, focus group discussions were held with care staff and family carers from five care homes in South London. Care home staff were recruited through the Maudsley Biomedical Research Centre Care Home Research Network in South London. Care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Focus groups for care staff and family carers were conducted separately. Following focus groups, an online survey was circulated to 16 family carers to specifically explore the themes that emerged from focus groups. 19 informal interviews were also conducted with night-time care staff and nurses. Thematic analysis revealed eight key themes in the management of sleep disturbance in people with dementia living in care homes: current night-time care practices, dissonance in perceived causes of sleep disturbances, inconsistencies in treatment options, insufficient staffing levels, working relationships between shifts, nurse burden and responsibility, communication as a critical challenge, connecting with residents and one overarching theme of balance. The key themes identified represent the major barriers to good quality care and areas which future programmes will need to address to improve the quality of night-time care in care homes (Nunez et al., Int J Geriatr Psychiatry. 2018;33(1):e140-e149). What was the aim of the study?

Select one:

- ☐ To evaluate the best interventions for sleep disturbances for people with dementia living in care homes
- ☐ To evaluate the barriers to care provision for people with dementia
- ☐ To clarify the challenges that exist for night-time care of people with dementia living in care homes
- ☐ To evaluate the causes of sleep disturbances for people with dementia living in care homes
- ☐ To evaluate the consequences of poor quality night-time care for people with dementia living in care homes

Your answer is incorrect.

In this study, the main aim is to identify the major barriers to good quality night-time care in care homes for people with dementia. The study did not intend to evaluate the consequences of poor quality care. The methods are not suited to evaluate interventions or causes of night-time behaviours.

The correct answer is: To evaluate the barriers to care provision for people with dementia

Question 90

Not answered

Marked out of 1.00

To explore the current practices and challenges in night-time care for people with dementia living in care homes in the UK, focus group discussions were held with care staff and family carers from five care homes in South London. Care home staff were recruited through the Maudsley Biomedical Research Centre Care Home Research Network in South London. Care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Focus groups for care staff and family carers were conducted separately. Following focus groups, an online survey was circulated to 16 family carers to specifically explore the themes that emerged from focus groups. 19 informal interviews were also conducted with night-time care staff and nurses. Thematic analysis revealed eight key themes in the management of sleep disturbance in people with dementia living in care homes: current night-time care practices, dissonance in perceived causes of sleep disturbances, inconsistencies in treatment options, insufficient staffing levels, working relationships between shifts, nurse burden and responsibility, communication as a critical challenge, connecting with residents and one overarching theme of balance. The key themes identified represent the major barriers to good quality care and areas which future programmes will need to address to improve the quality of night-time care in care homes (Nunez et al., Int J Geriatr Psychiatry. 2018;33(1):e140-e149). What was the sampling method used in this study?

Select one:

- ☐ Quota sampling
- ☐ Purposive sampling
- ☐ Consecutive sampling
- ☐ Cluster sampling
- ☐ Convenience sampling

Your answer is incorrect.

In this study, care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Thus purposive sampling, whereas selected individuals judged as suitable through a subjective nomination process are sampled.

The correct answer is: Purposive sampling

Question 91

Not answered

Marked out of 1.00

To explore the current practices and challenges in night-time care for people with dementia living in care homes in the UK, focus group discussions were held with care staff and family carers from five care homes in South London. Care home staff were recruited through the Maudsley Biomedical Research Centre Care Home Research Network in South London. Care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Focus groups for care staff and family carers were conducted separately. Following focus groups, an online survey was circulated to 16 family carers to specifically explore the themes that emerged from focus groups. 19 informal interviews were also conducted with night-time care staff and nurses. Thematic analysis revealed eight key themes in the management of sleep disturbance in people with dementia living in care homes: current night-time care practices, dissonance in perceived causes of sleep disturbances, inconsistencies in treatment options, insufficient staffing levels, working relationships between shifts, nurse burden and responsibility, communication as a critical challenge, connecting with residents and one overarching theme of balance. The key themes identified represent the major barriers to good quality care and areas which future programmes will need to address to improve the quality of night-time care in care homes (Nunez et al., Int J Geriatr Psychiatry. 2018;33(1):e140-e149). Based on the results reported here, which of the following inferences is the most appropriate?

Select one:

- ☐ There are areas of improvement in the care provided to people with dementia living in care homes
- ☐ More staff members are needed at day time to improve the provision of care at night in care homes
- ☐ Training specific to dementia care is required for staff at care homes
- ☐ Family members should be involved the provision of care at night in care homes
- ☐ Patients should be given regular online questionnaires to assess the quality of night time care they receive at care homes

Your answer is incorrect.

The most likely inference pertains to the several potential areas of improvement identified in the care provided to people with dementia living in care homes. This study did not evaluate the role of family members in night time care. There are no specific observations with respect to the staffing numbers during the day or dementia-related training.

The correct answer is: There are areas of improvement in the care provided to people with dementia living in care homes

Question 92

Not answered

Marked out of 1.00

To explore the current practices and challenges in night-time care for people with dementia living in care homes in the UK, focus group discussions were held with care staff and family carers from five care homes in South London. Care home staff were recruited through the Maudsley Biomedical Research Centre Care Home Research Network in South London. Care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Focus groups for care staff and family carers were conducted separately. Following focus groups, an online survey was circulated to 16 family carers to specifically explore the themes that emerged from focus groups. 19 informal interviews were also conducted with night-time care staff and nurses. Thematic analysis revealed eight key themes in the management of sleep disturbance in people with dementia living in care homes: current night-time care practices, dissonance in perceived causes of sleep disturbances, inconsistencies in treatment options, insufficient staffing levels, working relationships between shifts, nurse burden and responsibility, communication as a critical challenge, connecting with residents and one overarching theme of balance. The key themes identified represent the major barriers to good quality care and areas which future programmes will need to address to improve the quality of night-time care in care homes (Nunez et al., Int J Geriatr Psychiatry. 2018;33(1):e140-e149). Which of the following can be considered as a limitation of the reported study?

Select one:

- ☐ Use of the online survey instead of printed questionnaires
- ☐ Focussing on night-time care instead of overall care
- ☐ Using different qualitative approaches in the study
- ☐ Lack of patient interviews as a source of information
- ☐ Lack of reflection of the influence of the researcher on the results

Your answer is incorrect.

The use of different approaches such as focus group online survey as well as informal interviews has ensured a broad range of data to be available for thematic analysis. This is a particular strength. As the study question is about night-time care, it is essential that the focus be on the night-time care. The use of online survey in addition to interviews is likely to have increased participation, and thus another strength. There is no reflection of how the researchers' own experiences have shaped the thematic analysis and the subsequent results of this study. The lack of reflexivity is a limitation of this study.

The correct answer is: Lack of reflection of the influence of the researcher on the results

Question 93

Not answered

Marked out of 1.00

To explore the current practices and challenges in night-time care for people with dementia living in care homes in the UK, focus group discussions were held with care staff and family carers from five care homes in South London. Care home staff were recruited through the Maudsley Biomedical Research Centre Care Home Research Network in South London. Care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Focus groups for care staff and family carers were conducted separately. Following focus groups, an online survey was circulated to 16 family carers to specifically explore the themes that emerged from focus groups. 19 informal interviews were also conducted with night-time care staff and nurses. Thematic analysis revealed eight key themes in the management of sleep disturbance in people with dementia living in care homes: current night-time care practices, dissonance in perceived causes of sleep disturbances, inconsistencies in treatment options, insufficient staffing levels, working relationships between shifts, nurse burden and responsibility, communication as a critical challenge, connecting with residents and one overarching theme of balance. The key themes identified represent the major barriers to good quality care and areas which future programmes will need to address to improve the quality of night-time care in care homes (Nunez et al., Int J Geriatr Psychiatry. 2018;33(1):e140-e149). Which of the following can be considered as strength of the reported study?

Select one:

- ☐ Limiting sampling to 5 care homes in a specific region
- ☐ Using thematic analysis for the qualitative study
- ☐ Not using patient interviews as a source of information
- ☐ Focussing on night-time care instead of overall care
- ☐ Using different qualitative approaches to reach out to key groups

Your answer is incorrect.

The use of different approaches such as focus group online survey as well as informal interviews has ensured a broad range of data to be available for thematic analysis. This is a particular strength. As the study question is about night-time care, it is essential that the focus be on the night-time care. The sampling of 5 care homes only is not representative, and could be a limitation.

The correct answer is: Using different qualitative approaches to reach out to key groups

Question 94

Not answered

Marked out of 1.00

To explore the current practices and challenges in night-time care for people with dementia living in care homes in the UK, focus group discussions were held with care staff and family carers from five care homes in South London. Care home staff were recruited through the Maudsley Biomedical Research Centre Care Home Research Network in South London. Care home managers were asked to nominate members of staff to share their experiences of night-time care and sleep disturbance in people with dementia. Focus groups for care staff and family carers were conducted separately. Following focus groups, an online survey was circulated to 16 family carers to specifically explore the themes that emerged from focus groups. 19 informal interviews were also conducted with night-time care staff and nurses. Thematic analysis revealed eight key themes in the management of sleep disturbance in people with dementia living in care homes: current night-time care practices, dissonance in perceived causes of sleep disturbances, inconsistencies in treatment options, insufficient staffing levels, working relationships between shifts, nurse burden and responsibility, communication as a critical challenge, connecting with residents and one overarching theme of balance. The key themes identified represent the major barriers to good quality care and areas which future programmes will need to address to improve the quality of night-time care in care homes (Nunez et al., Int J Geriatr Psychiatry. 2018;33(1):e140-e149). In this study, focus groups for care staff and family carers were conducted separately. Which if the following is the most likely reason for this decision?

Select one:

- ☐ To avoid conflicts among family and staff carers
- ☐ To reduce interview costs
- ☐ To allow participants to freely express themselves
- ☐ To allow thematic analysis of the data
- ☐ To complete the study within the stipulated time

Your answer is incorrect.

For a focus group methodology to be productive, we may need to categorize our participants by relevant demographic or qualitative experience based variables whichever is relevant for the questions under consideration. This ensures that the groups are homogenous, and allows free expression of views. Ref: Smithson J. Using and analysing focus groups: limitations and possibilities. International journal of social research methodology. 2000 Jan 1;3(2):103-19.

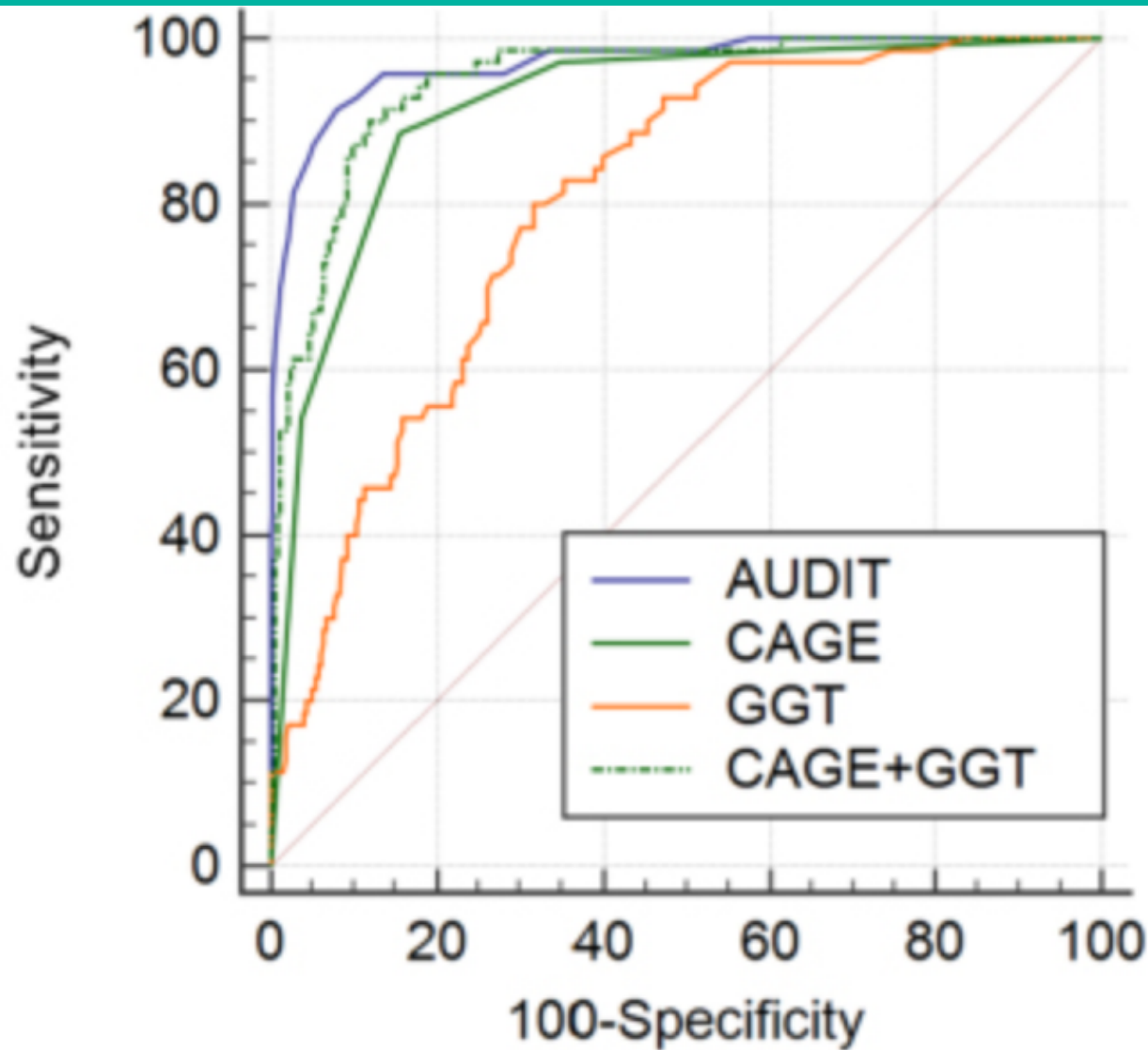
The correct answer is: To allow participants to freely express themselves

Question **95**

Not answered

Marked out of 1.00

Choe et al. tested the performance of the combined use of CAGE and GGT to screen alcohol dependence (AD). (Choe YM, et al. Neuropsychiatric Disease and Treatment. 2019;15:1507). Which of the following screening procedures is likely to have the largest area under the ROC curve?



(C) Non-AD vs. AD

Select one:

- ☐ CAGE + GGT
- ☐ AUDIT
- ☐ CAGE
- ☐ AUDIT + CAGE
- ☐ GGT

Your answer is incorrect.

The area under a ROC curve is larger for crives that are closer to the vertex than to the chance line. Among the given curves, the curve for AUDIT is the closest to the vertex and farthest from the chance line. This means that the discrimination accuracy for AUDIT is likely to be much better than that can be achieved by a random coin toss to diagnose alcohol dependence.

The correct answer is: AUDIT

Question **96**

Not answered

Marked out of 1.00

Choe et al. tested the performance of the combined use of CAGE and GGT to screen alcohol dependence (AD). (Choe YM, et al. Neuropsychiatric Disease and Treatment. 2019;15:1507). Which of the following screening procedures is likely to have the least discrimination accuracy for identifying alcohol dependence?